

THE **LEAN** SERIES

Jez Humble, Joanne Molesky & Barry O'Reilly

LEAN ENTERPRISE

Adopting Continuous
Delivery, DevOps, and
Lean Startup at Scale

Early Release

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Lean Enterprise

Jez Humble, Joanne Molesky, and Barry O'Reilly

LEAN ENTERPRISE

by Jez Humble, Joanne Molesky, and Barry O'Reilly

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Introduction

It's possible for good people, in perversely designed systems, to casually perpetrate acts of great harm on strangers, sometimes without ever realising it.

— BEN GOLDACRE

On April 1, 2010, California's only motor vehicle plant, New United Motor Manufacturing, Inc. (NUMMI) shut down. NUMMI, which opened in 1984, had been a joint venture between GM and Toyota. Both companies stood to benefit from the partnership. Toyota wanted to open a plant in the US to escape import restrictions threatened by the US Congress in reaction to the inexorably falling market share of US auto manufacturers. For GM, it was a chance to learn how to build small cars profitably, and to learn the Toyota Production System that has enabled Japanese auto manufacturers to consistently deliver the highest quality in the industry at costs that undercut those of US manufacturers.¹

For the joint venture, GM chose the site of their shuttered Fremont Assembly plant. GM's Fremont plant was one of their worst, both in terms of the quality of the cars produced, and the relationship between managers and workers. By the time the plant closed in 1982, labor relations had almost completely bro-

1. The story of the NUMMI plant is covered comprehensively in *This American Life*, episode 403: <http://www.thisamericanlife.org/radio-archives/episode/403/>, from which all the direct quotes are taken.

ken down, with workers drinking and gambling on the job. Incredibly, Toyota agreed to United Auto Worker negotiator Bruce Lee's demand to re-hire the union leaders from Fremont Assembly to lead the workforce at NUMMI. They were duly sent to Toyota City in Japan to learn the Toyota Production System. Within three months, the NUMMI plant was producing near perfect quality cars — some of the best quality in America, as good as the ones coming from Japan — at much lower cost than Fremont Assembly had achieved. Lee had been right in his bet that it was “it was the system that made it bad, not the people.”

Much has been written about the TPS (Toyota Production System), but a recurring theme can be heard when listening to the Fremont Assembly workers who ended up at NUMMI: teamwork. It might seem banal, but it was an incredibly powerful experience for many of the UAW employees. The TPS makes building quality in to products the highest priority, so problems must be fixed as soon as possible after they are discovered, and the system must then be improved to try and prevent them from happening again. Workers and managers cooperate to make this possible. The moment a worker discovers a problem, they can summon their manager by pulling on a cord (the famous *andon* cord). The manager will then come and help them try and resolve the problem. If the problem cannot be resolved within the time available, the worker can stop the production line to enable the problem to be fixed. The team will later experiment with, and implement, ideas to prevent the problem occurring again.

These ideas — that the primary task of managers is to help workers, that workers should have the power to stop the line, and that they should be involved in deciding how to improve the system — were revolutionary to the UAW employees. John Shook, the first American to work in Toyota City, who had the job of training the NUMMI workers, reflects that “they had had such a powerful emotional experience of learning a new way of working, a way that people could actually work together collaboratively — as a team.”

The way the TPS works is in sharp contrast to traditional US and European management practice, which is based on the principles of Frederick Winslow Taylor, the creator of Scientific Management. According to Taylor, the job of management is to analyze the work and break it down into discrete tasks. These tasks are then performed by specialized workers who need understand nothing more than how to perform their particular specialized task as efficiently as possible. Taylorism fundamentally thinks of organizations as machines which are to be analyzed and understood by breaking them down into their component parts.

In contrast, the heart of the Toyota Production System is the creation of a high-trust culture in which everybody is aligned in their goal of building a high-quality product on demand, and where workers and managers collaborate across functions to constantly improve — and sometimes radically redesign — the system. These ideas from the Toyota Production System — a high-trust culture focused on continuous improvement (*kaizen*), powered by high levels of both alignment and autonomy at all levels, are essential to building a large organization that can adapt rapidly to changing conditions.

A key part of the success of the TPS can be found in its effect on workers. Taylorism demands workers that are more or less cogs in a machine, paid simply to perform pre-planned actions as quickly as possible. The TPS, instead, requires workers to pursue mastery through continuous improvement, imbues them with a higher purpose — the pursuit of ever-higher levels of quality, value, and customer service — and provides a level of autonomy by empowering them to experiment with ideas for improvement. Autonomy, mastery and purpose are the intrinsic motivators which Dan Pink shows are the keys to creating a highly motivated, fulfilled workforce (as opposed to extrinsic motivators such as salary or share options). Rick Madrid, who worked at the Fremont plant both before and during the NUMMI era, says of the Toyota Production System that “it changed my life from being depressed, bored — and like my son said, it changed my attitude. It changed me all for the better.”

Although the principles at the heart of the TPS might seem relatively straightforward, they are very hard to adopt. Indeed, GM utterly failed in taking what it had achieved at NUMMI and reproducing it in other GM plants. Some of the biggest obstacles were changes to the organizational hierarchy. The TPS does away with the concept of seniority, in which union workers are assigned jobs based on how many years of service they have, with the best jobs going to the most senior. Under the TPS, everybody has to learn all the jobs required of their team, and rotate through them. The TPS also removes the visible trappings and privileges of management. Nobody wore a tie in the NUMMI plant — not even contractors — to emphasize the fact that everybody was part of the same team. Managers did not receive perks accorded them at other GM plants, such as a separate cafeteria and car park.

Finally, attempts to improve quality ran up against organizational boundaries. In the TPS, suppliers, engineers, and workers collaborate to continuously improve the quality of the parts and to make sure that workers have the tools they

need to do their job. This worked at NUMMI because the engineers were in-house, and the parts came from Japanese suppliers who had a collaborative relationship with Toyota. In the US supply chain, things were different. If the parts that came in to GM assembly plants were of poor quality, or didn't fit, there was simply no mechanism to fix the problem.

Ernie Schaefer, manager of GM's Van Nuys plant — which faced many of the same problems as Fremont Assembly — describes what was different about NUMMI. He says, “you can see a lot of things different. But the one thing you don't see is the system that supports the NUMMI plant. I don't think, at that time, anybody understood the large nature of this system. General Motors was a kind of throw it over the wall organization. Each department, we were very compartmentalized, and you design that vehicle, and you'd throw it over the wall to the manufacturing guys.” This is the legacy of a Taylorist management approach. The TPS exists — and can only succeed — within an ecosystem of organizational culture, supplier relations, financial management, HR and governance that is designed around its philosophy.

GM tried to implement the TPS at Van Nuys, but failed. Workers and managers rebelled in the face of changes in status and behavior that were required of them, despite the threat of closure (which was ultimately carried out). According to Larry Spiegel, a veteran of NUMMI who had been sent to Van Nuys to help implement the TPS, people at the plant simply didn't believe the threats to shut it down. He says, “There were too many people convinced that they didn't need to have to change.”

This lack of urgency acted as a barrier to adoption across GM (and is perhaps the biggest obstacle to organizational change in general).² The US division of GM took about fifteen years to decide they needed to seriously prioritize implementing the TPS, and a further ten years to actually implement it. By this time any competitive advantage they could have gained in the process was lost. GM went bankrupt and was bailed out by the US government in 2009, at which point it pulled out of NUMMI. Toyota shut down the NUMMI plant in 2010.

2. John Kotter, author of *Leading Change*, says, “a majority of employees, perhaps 75 percent of management overall, and virtually all of the top executives need to believe that considerable change is absolutely essential” ((to come), p51)

The story of NUMMI is important because it illustrates the main concern of this book — growing a lean enterprise, such as Toyota — and many of the common obstacles. Toyota has always been very open about what it is doing, giving public tours of its plants, even to competitors — partly because it knows that what makes the TPS work is not so much what is visible, including the particular practices, but the culture. Many people focus on the practices and tools popularized by the TPS, such as andon cords and kanban system. One GM vice president even ordered one of his managers to take pictures of every inch of the NUMMI plant so they could copy it precisely. The result: a factory with andon cords, but nobody pulling them because managers were incentivized by the rate at which cars (of any quality) came off the line.

A Lean Enterprise is Primarily a Human System

As the pace of social and technological change in the world accelerates, the lean approach pioneered by Toyota becomes ever more important because it sets out a proven strategy for thriving in uncertainty through embracing change. The key to understanding Lean Thinking is that it is primarily a *human* system. It is common for people to focus on specific practices and tools that lean and agile teams use such as kanban boards, stand-up meetings, pair programming and so forth. However too often these are adopted as rituals or “best practice” rather than being seen for what they really are — *countermeasures* that are effective within a particular context in the pursuit of a particular goal.

In an organization with a culture of continuous improvement, these countermeasures emerge naturally within teams and are then discarded when they are no longer valuable. The key to creating a lean enterprise is to enable the people doing the work to solve their customers’ problems in a way that is aligned with the strategy of the wider organization. In order to achieve this we rely on people being able to make decisions locally that are sound at a strategic level, which in turn, relies critically on the flow of information, including feedback loops.

Information flow has been studied extensively by Ron Westrum, primarily in the context of accidents and human error in aviation and healthcare. Westrum realized that safety in these contexts could be predicted by organizational culture, and developed a “continuum of safety cultures” with three categories: ³

3. (to come)

Pathological organizations are characterized by large amounts of fear and threat. People often hoard information or withhold it for political reasons, or distort it to make themselves look better. Bureaucratic organizations protect departments. Those in the department want to maintain their “turf,” insist on their own rules, and generally do things by the book — their book. Generative organizations focus on the mission. How do we accomplish our goal? Everything is subordinated to good performance, to doing what we are supposed to do.

These cultures process information in different ways. Westrum observes that “the climate that provides good information flow is likely to support and encourage other kinds of cooperative and mission-enhancing behaviour, such as problem solving, innovations, and inter-departmental bridging. When things go wrong, pathological climates encourage finding a scapegoat, bureaucratic organizations seek justice, and the generative organization tries to discover the basic problems with the system.” The characteristics of the various types of culture are shown in [Table 1-1](#).

Table 1-1. *How Organizations Process Information*

Pathological (<i>power oriented</i>)	Bureaucratic (<i>rule oriented</i>)	Generative (<i>performance oriented</i>)
Low cooperation	Modest cooperation	High cooperation
Messengers shot	Messengers neglected	Messengers trained
Responsibilities shirked	Narrow responsibilities	Risks are shared
Bridging discouraged	Bridging tolerated	Bridging encouraged
Failure leads to scapegoating	Failure leads to justice	Failure leads to enquiry
Novelty crushed	Novelty leads to problems	Novelty implemented

Westrum’s typology has been extensively elaborated upon, and has a visceral quality that will appeal to anybody who has worked in a pathological (or even bureaucratic) organization. However it has implications that are far from academic.

In 2013, PuppetLabs, IT Revolution Press and ThoughtWorks surveyed 9,200 technologists worldwide to find out what made high-performing organizations successful. The resulting *2014 State of DevOps Report* is based on analysis of

answers from people working in a variety of industries including finance, telecoms, retail, government, technology, education and healthcare. The headline result from the survey was that strong IT performance is a competitive advantage. Analysis showed that firms with high-performing IT organizations were *twice as likely* to exceed their profitability, market share and productivity goals.

4

The survey also set out to examine the cultural factors that influenced organizational performance. The most important of these turned out to be whether people were satisfied with their jobs, based on the extent to which they agreed with the following statements — which are strongly reminiscent of the reaction of the NUMMI workers who were introduced to the Toyota Production System:

- I would recommend this organization as a good place to work.
- I have the tools and resources to do my job well.
- I am satisfied with my job.
- My job makes good use of my skills and abilities.

The fact that job satisfaction was the top predictor of organizational performance strongly supports Pink’s argument on the importance of intrinsic motivation. The team working on the survey wanted to look at whether Westrum’s model was a useful tool to predict organizational performance.⁵ Thus the survey asked people to assess their team culture across each of the axes of Westrum’s model as shown in [Table 1-1](#), by asking them to rate the extent to which they agreed with statements such as “On my team, failure causes enquiry.”⁶ In this way, the survey was able to *measure* culture.

4. The survey measured organizational performance by asking respondents to rate their organization’s relative performance in terms of achieving its profitability, market share and productivity goals. This is a standard scale that has been validated multiple times in prior research. See (to come).
5. In the interests of full disclosure, Jez was part of the team behind the *2014 State of DevOps Report*.
6. This method of measuring attitudes quantitatively is known as a Likert scale.

Statistical analysis of the results showed that team culture was not only strongly correlated with organizational performance; it was also a strong predictor of job satisfaction. The results are clear: a high-trust, generative culture is not just important to creating a safe working environment — it is the foundation of creating a high performance organization.

Mission Command: An Alternative to Command and Control

High-trust organizational culture is often contrasted to what is popularly known as “command and control” — the idea from Scientific Management that the people in charge make the plans and the people on the ground execute them — which is usually thought to be modelled on the way the military functions. In reality, however, this type of command and control has not been fashionable in military circles since 1806 when the Prussian army, a classic plan-driven organization, was decisively defeated by Napoleon’s decentralized, highly motivated forces. Napoleon used a style of war known as *maneuver warfare* to defeat larger, better trained armies. In maneuver warfare, the goal is to minimize the need for actual fighting by instead disrupting your enemy’s ability to act cohesively through the use of shock and surprise. A key element in maneuver warfare is being able to learn, make decisions, and act faster than your enemy — the same capability that allows startups to disrupt enterprises.⁷

Three men are especially important to the reconstruction of the Prussian army following its defeat by Napoleon’s armies: Carl von Clausewitz, David Scharnhorst, and Helmuth von Moltke. Their contributions not only transformed military doctrine, they have important implications for people leading and managing large organizations — particularly the doctrine of *Auftragstaktik*, or Mission Command, which we will explore here. Mission Command is what enables maneuver warfare to work at scale, and is key to understanding how enterprises can compete with startups.

Following the eventual defeat of Napoleon, General David Scharnhorst was made Chief of the newly-established Prussian General Staff. He put together a reform commission which conducted a post-mortem and then began to trans-

7. As we discuss in [Chapter 3](#), this concept is formalized in John Boyd’s OODA (observe-orient-decide-act) loop, which in turn inspired Eric Ries’ build-measure-learn loop.

form the Prussian Army. Scharnhorst noted that Napoleon's officers had the authority to make decisions as the situation changed on the ground without waiting for approval through the chain of command, allowing them to adapt rapidly to changing circumstances.

Scharnhorst wanted to develop a similar capability in a systematic way, and realized this required the training of a independent, intelligent cadre of staff officers who shared similar values and would be able to act decisively and autonomously in the heat of battle. Thus military schools were set up to train staff officers, who for the first time were accepted from all social backgrounds based on merit.

In 1857 Helmuth von Moltke, perhaps best known for his saying "no plan survives contact with the enemy," was appointed Chief of the General Staff of the Prussian Army. His key innovation, building on the military culture established by Scharnhorst, was to treat military strategy as a series of options which were to be explored extensively by officers in advance of the battle. In 1869 he issued a directive titled "Guidance for Large Unit Commanders", which sets out how to lead a large organization under conditions of uncertainty.

In this document, von Moltke notes that "In war, circumstances change very rapidly, and it is rare indeed for directions which cover a long period of time in a lot of detail to be fully carried out." He thus recommends "not commanding more than is strictly necessary, nor planning beyond the circumstances you can foresee." Instead, he has this advice: "The higher the level of command, the shorter and more general the orders should be. The next level down should add whatever further specification it feels to be necessary, and the details of execution are left to verbal instructions or perhaps a word of command. This ensures that everyone retains freedom of movement and decision within the bounds of their authority... The rule to follow is that an order should contain all, but also only, what subordinates cannot determine for themselves to achieve a particular purpose."

Crucially, orders always include a passage which describes their *intent*, communicating the *purpose* of the orders. This allows subordinates to make good decisions in the face of emerging opportunities or obstacles which prevent them following the original orders exactly. Von Motke notes that "there are numerous situations in which an officer must act on his own judgment. For an officer to wait for orders at times when none can be given would be quite absurd. But

as a rule, it is when he acts in line with the will of his superior that he can most effectively play his part in the whole scheme of things.”

These ideas form the core of the doctrine of *Auftragstaktik*, or Mission Command, which, in combination with the creation of a professionally-trained cadre of staff officers who understood how to apply the doctrine operationally, was adopted by multiple elite military units (including the US Marine Corps) as well as (more recently) NATO.

The history of the Prussian Army’s development of *Auftragstaktik* is described in more detail in Stephen Bungay’s treatise on business strategy, *The Art of Action* (from which the quotations above from “Guidance for Large Unit Commanders” are taken). Bungay develops a theory on the problems of directing strategy at scale which builds on the work of Scharnhorst, von Moltke, and another Prussian general: Carl von Clausewitz. As a 26-year old, Clausewitz had fought against Napoleon in the fateful battles of Jena and Auerstadt. He subsequently served on Scharnhorst’s reform commission, and bequeathed us his unfinished *magnum opus*, *On War*. In this work he introduces the concept of the “fog of war” — the fundamental uncertainty we face as actors in a large and rapidly changing environment, with necessarily incomplete knowledge of the state of the system as a whole. He also introduces the idea of *friction*, that which prevents reality from behaving in an ideal way. Friction exhibits itself in the form of incomplete information, unanticipated side-effects, human factors such as mistakes and misunderstandings, and the accumulation of unexpected events.

Friction and Complex Adaptive Systems

Clausewitz’ concept of friction is an excellent metaphor to understand the behavior of complex adaptive systems, such as an enterprise (or indeed any human organization). The defining characteristic of a complex adaptive system is that its behavior at a global level cannot be understood through Taylor’s reductionist approach of analyzing its component parts. Rather, many properties and patterns of behavior of complex adaptive systems at the global level “emerge” from *interactions* between events and components at multiple levels within the system. In the case of open systems (such as enterprises) we also have to consider interactions with the environment, including the actions of customers and competitors, as well

as wider social and technological changes.⁸ Friction is ultimately a consequence of the human condition — the fact that organizations are composed of people with independent wills and limited information. Thus friction cannot be overcome.

Bungay argues that friction creates three gaps. First, a *knowledge gap* that arises when engaged in planning or acting due to the necessarily imperfect state of the information we have to hand, and our need to making assumptions and interpret that information. Second, an *alignment gap* which is the result of people failing to do things as planned, perhaps due to conflicting priorities, misunderstandings, or simply someone forgetting or ignoring some element of the plan. Finally, there is an *effects gap* due to unpredictable changes in the environment, perhaps due to other actors, or unexpected side-effects which create outcomes that differ from those we anticipated. These gaps are shown in [Figure 1-1](#).

8. For those interested in different types of systems and how to make sense of them, we recommend studying Dave Snowden's Cynefin framework: <http://www.youtube.com/watch?v=N7oz366X0-8>

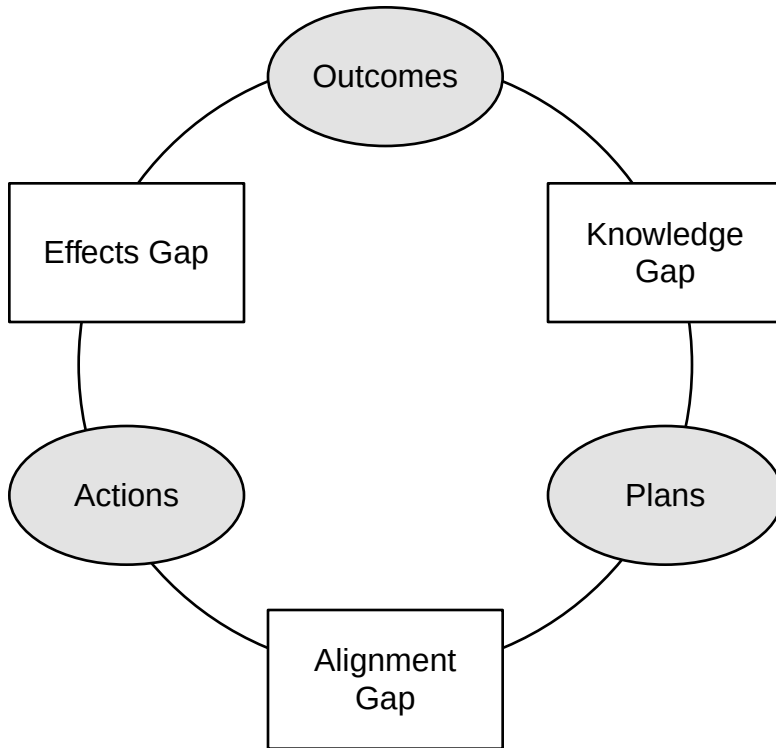


Figure 1-1. *Gaps in complex adaptive systems.*

Bungay then goes on to describe the the usual Scientific Management remedy applied by enterprises, the alternative proposed by the doctrine of *Auftragstaktik*, and his own interpretation of Mission Command as applied to business, which he terms “directed opportunism”. These are shown in [Table 1-2](#).

Table 1-2. *The three gaps, and how to manage them.*

	Effects Gap	Knowledge Gap	Alignment Gap
What is it?	The difference between what we expect our actions to achieve and what they actually achieve	The difference between what we would like to know and what we actually know	The difference between what we want people to do and what they actually do

	Effects Gap	Knowledge Gap	Alignment Gap
Scientific Management remedy	More detailed controls	More detailed information	More detailed instructions
<i>Auftragstaktik</i> remedy	"Everyone retains freedom of decision and action within bounds"	"Do not command more than is necessary or plan beyond the circumstances you can foresee"	"Communicate to every unit as much of the higher intent as is necessary to achieve the purpose"
Directed Opportunism remedy	Give individuals freedom to adjust their actions in line with intent	Limit direction to defining and communicating the intent	Allow each level to define how they will achieve the intent of the next level up, and 'backbrief'

What is crucial to understand is that when we are working in a complex adaptive system, where friction dominates, the Scientific Management remedies *cannot work*. In fact, they make things worse. Creating ever more detailed plans delays the feedback which tells us which of our assumptions are invalid. Complex sets of rules and controls punish the innocent but can be evaded by the guilty, all the while destroying morale, innovation, and entrepreneurialism. Intelligence gathering fails in the face of bureaucratic or pathological organizations which hide or distort information in order to protect their turf. Organizations which cannot escape the grip of Scientific Management are perfect targets for disruption by organizations which understand how to move fast at scale.

Create Alignment At Scale Following the Principle of Mission

The most important concern leaders and managers operating within a complex adaptive system face is this: how can we enable people within the organization to make good decisions — act in the best interests of the organization — given that they can *never* have sufficient information and context to understand the full consequences of their decisions, and given that events often overtake our plans?

In *The Principles of Product Development Flow*, Donald Reinertsen presents the *Principle of Mission*, based on the doctrine of mission command, in which we “specify the end state, its purpose, and the minimum possible constraints”. According to the Principle of Mission we create alignment not by providing a detailed plan of how we achieve our objective, but by describing the *intent* of our mission and communicating *why* we are undertaking it.

The key to the Principle of Mission is to create alignment, and enable autonomy, by setting out clear, high-level target conditions with an agreed time-frame — which gets smaller under conditions of greater uncertainty — and then leaving the details of *how* to achieve the conditions to teams. This approach can even be applied to multiple levels of hierarchy, with each level reducing the scope while providing more context. In the course of the book, this principle is applied in multiple contexts:

- **Budgeting and financial management:** Instead of a traditional budgeting process which requires all spending for the next year to be planned and locked down based on detailed projections and business plans, we set out high level objectives across multiple perspectives such as people and organization, operations, market, and finance that are reviewed regularly. This kind of exercise can be used at multiple levels, with resources being allocated dynamically based upon need, and the indicators to be used reviewed on a regular basis.
- **Program management:** Instead of creating detailed, up-front plans on the work to be done and then breaking work down into tiny little bits which are then distributed to individual teams, we specify at the program level only the measurable objectives for each iteration. The teams then to work out how to achieve those objectives, including collaborating with other teams and continuously integrating their work to ensure they will meet the program-level objectives.
- **Process improvement:** Working to continuously improve processes is a key element of the Toyota Production System, as well as powerful tool to transform organizations. In (to come) we present the Improvement Kata, in which we work in iterations, specifying target objectives for processes, and providing the people operating the processes the time and resources to run experiments to try and meet the target objectives for the next iteration.

Crucially these mission-based processes must *replace* the command and control processes, rather than running alongside them. This requires people to behave and act in different ways and learn new skills. It also requires cultural change within the organization, as we discuss in (to come). Discussing how to apply mission command in business, Stephen Bungay reflects on the culture that enables mission command — which, through no coincidence, has the same charac-

teristics that we find in the generative organizations described by Westrum in [Table 1-1](#).

*The unchanging core is a holistic approach which affects recruiting, training, planning and control processes, but also the culture and values of an organization. Mission command embraces a conception of leadership which unsentimentally places human beings at its center. It crucially depends on factors which do not appear on the balance sheet of an organization: the willingness of people to accept responsibility; the readiness of their superiors to back up their decisions; the tolerance of mistakes made in good faith. Designed for an external environment which is unpredictable and hostile, it builds on an internal environment which is predictable and supportive. At its heart is a network of trust binding people together up, down, and across a hierarchy. Achieving and maintaining that requires constant work.*⁹

The Ability To Innovate Is Your Competitive Advantage

The story of the Fremont Assembly site doesn't stop with NUMMI. It is in fact the locus of two paradigm shifts in the US auto manufacturing industry. In 2010, the NUMMI plant was purchased by Tesla Motors, and became the Tesla Factory. Tesla is using continuous methods to innovate faster than Toyota, discarding the concept of model years in favor of more frequent updates, and in many cases enabling owners of older cars to download new firmware to gain access to new features. Tesla has also championed transparency of information, announcing it will not enforce its patents. In doing so, it echoes a story from Toyota's origins building automatic looms. Upon hearing that the plans for one of the looms had been stolen, Kiichiro Toyoda is said to have remarked,

9. (to come), p88.

*Certainly the thieves may be able to follow the design plans and produce a loom. But we are modifying and improving our looms every day. So by the time the thieves have produced a loom from the plans they stole, we will have already advanced well beyond that point. And because they do not have the expertise gained from the failures it took to produce the original, they will waste a great deal more time than us as they move to improve their loom. We need not be concerned about what happened. We need only continue as always, making our improvements.*¹⁰

The long-term value of an enterprise is not captured by the value of its products and intellectual property, but rather its ability to continuously increase the value it provides to customers — and to create *new* customers — through innovation.

A key premise of this book — supported by the experience of companies such as Tesla among many, many others — is that by putting IT and software development at the heart of your organization, you can accelerate the innovation cycle, providing your enterprise with a competitive advantage. This is supported by the data from the “2014 State of DevOps Report.” IT is a competitive advantage because it allows you to search for new opportunities and execute existing opportunities faster than the competition. The good news is that these capabilities are within the reach of all enterprises, not just tech giants. The data from the report shows that 20% organizations with more than 10,000 employees fall into the high performing group — a smaller percentage than smaller companies, but still significant. The purpose of this chapter is to show that organizational culture and strategy are the critical factors.

Many organizations try to take short-cuts to higher performance by starting innovation labs, acquiring startups, adopting methodologies, or through reorganizations. But such efforts are neither necessary nor sufficient. They can only succeed if combined with efforts to create a generative culture and strategy across the whole organization, including suppliers — and if this is achieved, there will be no need to resort to such short-cuts.

(Brief introduction to rest of book)

10. (to come), p40, emphasis ours.

Manage the Dynamics of the Enterprise Portfolio

The purpose of a business is to create a customer.

— PETER DRUCKER

In this chapter we will examine the lifecycle of businesses, and how companies can balance the *exploration* of new business models with the *exploitation* of proven ones. We'll need this distinction in order to understand where lean start-up practices and principles can be applied in an enterprise context, and how they can be used as the basis of managing an innovation portfolio.

In his book *Diffusion of Innovations*, Everett Rogers describes the cycle through which all successful technologies and ideas progress, shown in [Figure 2-1](#). Over time, all successful ideas, whether technologies, product categories, business models, or even methodologies, progress from being scarce and unevenly distributed to eventually becoming a commodity. They then form the building blocks for new, higher-level, more valuable innovations. Of course the time it takes for innovations to progress through the various stages of the cycle can vary substantially.

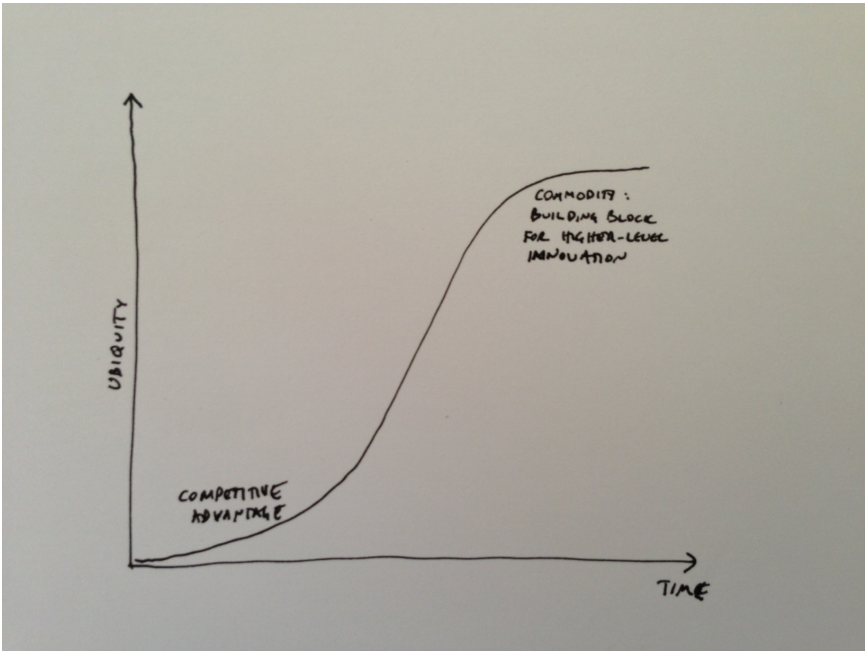


Figure 2-1. *The S-curve which shows the lifecycle of innovations.*

Rogers believed people could be classified into groups based on how they respond to innovation, as shown in [Figure 2-2](#). Initially new technologies and ideas are experimented with and tested by innovators, which form the smallest group of the overall population. As innovators discover technologies or ideas which provide competitive advantage — most will not — they will be taken up by the early adopters, and in this way success in each group leads to further diffusion through the other groups. Rogers' ideas were popularized and built upon by Geoffrey Moore, who introduced the concept of the “chasm”, as a logical divide between uptake by early adopters and the early majority. This chasm was inspired by Moore's observation that many innovations flounder once they are no longer seen as a source of competitive advantage by visionaries, but have not yet been widely enough adopted to be seen as a safe bet or proven practice by people in the early majority.

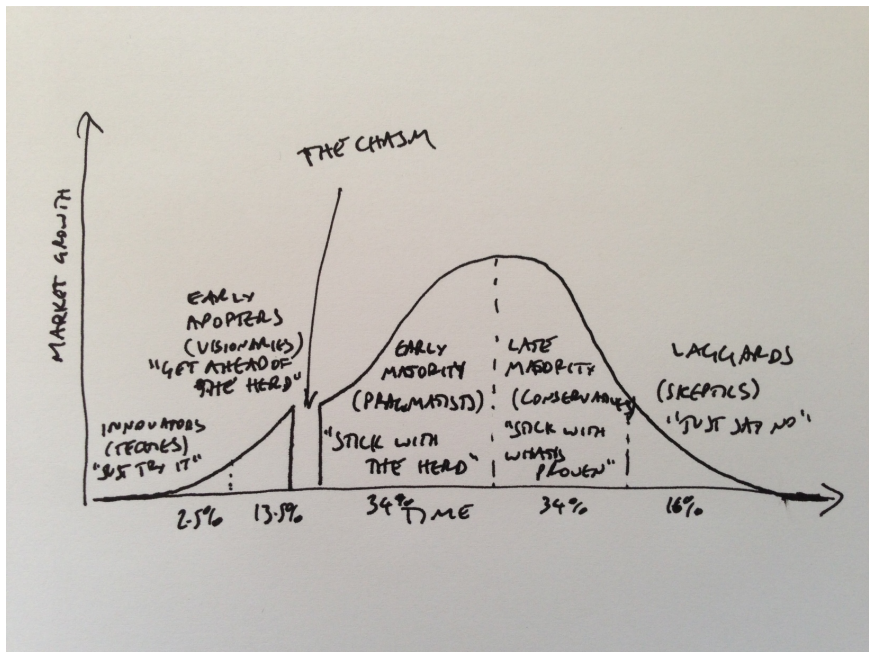


Figure 2-2. Technology adoption lifecycle.

Once a disruptive new technology or idea has been assimilated by the market a whole range of product offerings are spawned. Moore's take on the product category lifecycle is shown in [Figure 2-3](#). A successful product category will initially see high growth (stage B), following which it will become a mature market (stage C) in which consolidation takes place. Growth in mature markets is typically driven by acquiring competitors and new customers, and through efficiency gains. Finally, product categories decline (stage D). At any point a category can be disrupted through some new innovation — indeed an innovation is defined as “disruptive” based on its effect on existing product categories and business models. Even in the face of disruption, it's sometimes possible to maintain a lucrative niche market; for instance feature phones are still an important category in many countries, and IBM is still has a profitable main-frame business.

The Category-Maturity Life Cycle

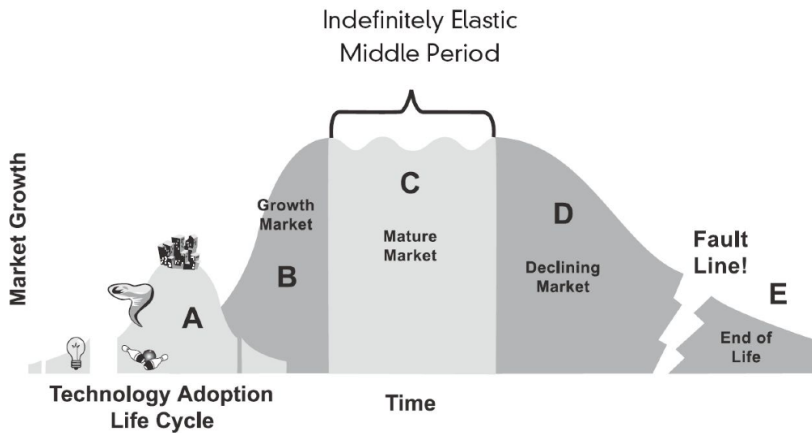


Figure 2-3. Category maturity lifecycle.

The first point to observe is that products in different stages of maturity vary significantly in terms of how they are managed, developed, marketed, and funded. For example, in a mature market we have a good understanding of our customers and the value the product provides them, the process of acquiring new customers or selling new products to existing ones is known, and new products in the category typically contain only incremental innovations compared to existing ones. For new categories, the opposite is true.

While there is a lot of detail involved in understanding these different stages in the lifecycle, including other dimensions such as whether our customers are other businesses or consumers, we can drive out most of the important discussion by drastically simplifying the problem and looking at two key activities that all enterprises will engage in: *exploring* new product categories and business models, and *exploiting* proven ones.¹

Startups begin by exploring new opportunities through business model innovation: they search for a new business model which is aligned with the founders'

1. This distinction was first proposed by James March in his paper "Exploration and Exploitation in Organizational Learning" ((to come)). Steve Blank refers to these activities as "search" and "execute" in the context of customer development.

vision, delivers value for customers, and which can drive the profitability and growth of the organization. Once this has been achieved, the business model is *exploited* through growing and scaling it, and finding ways to drive down costs, improve efficiency and increase market share and customer base. However every business model is ultimately transient, and eventually every business model and product category will be disrupted by new ones — it is only a matter of time.

Exploring new opportunities and exploiting existing ones are fundamentally different strategies, requiring different structure, competencies, processes, and mindset. It is hard to overemphasize this key point: management practices that are effective in the exploit domain will lead to failure if applied to exploring new opportunities — and vice-versa. The differences between these two domains are listed in [Table 2-1](#).

Table 2-1. *Explore versus Exploit*

	Explore	Exploit
Strategy	Radical or disruptive innovation, new business model innovation	Incremental innovation, existing business model optimization
Structure	Small cross-functional multi-skilled team	Multiple teams aligned using Principle of Mission
Culture	High tolerance for experimentation, risk taking, acceptance of failure, focus on learning	Incrementally improve and optimize, values quality and customer satisfaction
Risk management	Biggest risk is failure to achieve product/market fit	A more complex set of trade-offs specific to each product/service
Goals	Create new markets, discover new opportunities within existing markets	Maximize yield from captured market, outperform competitors
Measure of progress	Achieving product/market fit	Outperform forecasts, achievement of planned milestones and targets

Startups which discover a successful business model and cross the chasm often find it hard to transition into the next stage: executing and scaling in a growth market. Meanwhile, organizations that succeed in transforming themselves into engines of execution often lose their ability to explore new business models. Eric Ries wrote himself an imaginary memo capturing this shift in mindset:

Dear Eric, thank you for your service to this company. Unfortunately, the job you have been doing is no longer available, and the company you used to work for no longer exists. However, we are pleased to offer you a new job at an entirely new company, that happens to contain all the same people as before. This new job began months ago, and you are already failing at it. Luckily, all the strategies you've developed that made you successful at the old company are entirely obsolete. Best of luck! ²

A key goal of successful portfolio management in an enterprise is to understand how to balance exploring new businesses with exploiting proven existing business models — along with how to transition businesses successfully from one state to the other. Equally, leaders must understand the difference between these two domains, and be able to operationalize the very different mindsets and strategies that govern them.

Exploring New Ideas

Less than 50% of startups are alive five years after they start. ³ In a similar way, enterprises waste enormous amounts of money on trying to grow new businesses, creating little to no value for customers. ⁴ Of course it's impossible to know in advance whether or not a new business will be successful, but Eric Ries' *The Lean Startup* details a method for proceeding in conditions of extreme uncertainty. The Lean Startup methodology applies within the enterprise context just as it does in the world of startups, so long as we are clear on its purpose: *to discover and operationalize new and potentially disruptive business models*.

Every entrepreneur, whether they work in a startup or an enterprise, has a vision of their business and the impact it will have on legions of grateful, adoring customers. In order for this vision to become reality, there are two key assump-

2. <http://www.startuplessonslearned.com/2008/11/what-is-customer-development.html>
3. <http://smallbiztrends.com/2012/12/start-up-failure-rates-the-definitive-numbers.html> — these numbers vary by industry, with the 5 year survival rate for infotech businesses being substantially lower than education and health: <http://smallbiztrends.com/2008/05/startup-failure-rates-vary-across-economy.html>
4. Hard data is harder to come by, but circumstantial evidence is everywhere

tions that must be tested: the *value* hypothesis and the *growth* hypothesis. The value hypothesis asks whether our business actually provides value for users by solving a real problem. If so, we can say we have found *problem/solution fit*. The growth hypothesis tests how fast we can acquire new customers, demonstrating that we have a repeatable customer development process — in other words, that our customer base will move rapidly up the “hockey stick” in (to come), with a sufficiently low customer acquisition cost. If we pass these tests, we have *product/market fit*, and we can proceed to the final two stages in Steve Blank’s customer development process: *customer creation*, in which we launch our business in earnest, followed by *company building* where we attempt to cross the chasm.⁵

In the Lean Startup methodology, we take a systematic approach to working through this process iteratively. We start by working out what we need to *learn* by creating a value hypothesis. We then decide what to *measure* in order to test that hypothesis. We then design an experiment, called the *minimum viable product*, which we *build* in order to gather the necessary data from real customers to determine if we have product/market fit.

The trick is to invest the minimum amount of work to go through this cycle. Because we are operating in conditions of extreme uncertainty, we expect that our value hypothesis will be incorrect. At this point we *pivot*, coming up with a new value hypothesis based on our learning and then go through the process again. We keep doing this until we either achieve product/market fit, decide to stop experimenting or funding dries up. The amount of time we have before the money runs out is known as the *runway*, and the goal is to pivot as frequently as possible in order to find product/market fit while we still have runway left.

An important characteristic of the Lean Startup method is that experiments are cheap and quick to run relative to building out a complete product. In general we are able to build out a minimum viable product and gather data in the space of hours, days or weeks rather than months or years, using small, cross-functional teams that are focused on executing the build-measure-learn feedback loop shown in [Figure 2-4](#). Because it’s relatively cheap to explore new ideas, we can optimize for cycle time of testing our ideas in order to try out lots of them. We

5. [Link to reference for Steve Blank’s CusDev process](#)

expect many experiments will fail, and that a few will succeed but all will result in validated learning.

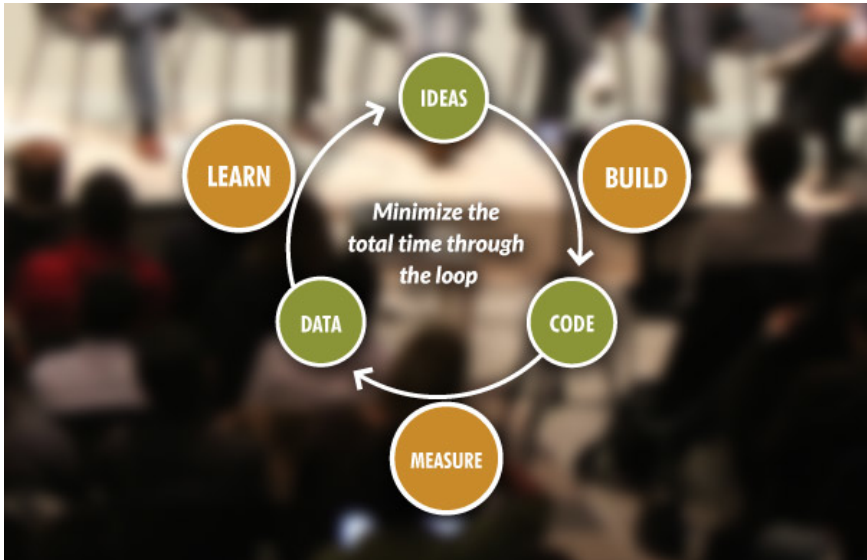


Figure 2-4. *The Build-Measure-Learn Loop*

In an Enterprise context, because the Lean Startup is relatively cheap to run, we can pursue multiple possible business models simultaneously using the principle of *optionality*.

NOTE

What is an Option?

Purchasing an option gives us the right, but no obligation, to do something in the future (typically to buy or sell an asset at a fixed price). Options have a price and an expiry date. Concert tickets, an agreement to go out for dinner with someone, and a decision to fund a product or service are all examples of options.

Investing a fixed amount of money and time to investigate the economic parameters of an idea — whether a business model, product, project, or an innovation such as a process change — is an example of using optionality to manage the uncertainties around the decision to invest further. The idea is to limit our maximum investment loss (“downside”) on any individual idea, with the expectation that a small number of ideas or decisions will pay off big time, and

offset or negate investments in those that did not, as shown in Figure 2-5.⁶ Optionality is a powerful concept that lets us defer decisions on how to achieve a desired outcome by exploring multiple possible approaches simultaneously.

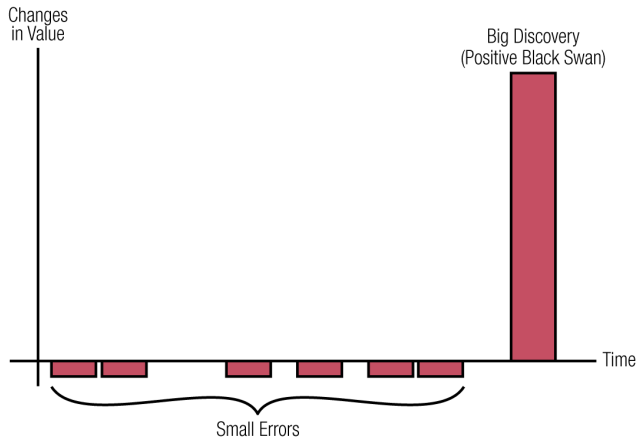


Figure 2-5. *The Principle of Optionality*

TIP

Effectuation

Limiting downside and making sure every decision creates at least *some* upside (even if it is just new information) is one of several techniques that entrepreneurs apply in situations of uncertainty, where simple cause-effect reasoning is an inappropriate way to manage risk. In her book *Effectuation: Elements of Entrepreneurial Expertise*, cognitive scientist Dr Saras Sarasvathy describes a framework for entrepreneurship based on research into how entrepreneurs work in real life.⁷

6. This idea of optionlike trial and error, or tinkering, is explored in Nassim Taleb's *Antifragile*, p181ff. Using the language of Dave Snowden's Cynefin framework, options are a way to make experiments "safe to fail", meaning that they are designed in such a way that we have limited the possible negative outcomes associated with failure. For more on the application of options to IT management, read *Commitment*, by Olav Maassen, Chris Matts and Chris Geary.
7. For a 3-page guide to the effectuation framework, visit <http://www.effectuation.org/sites/default/files/documents/effectuation-3-pager.pdf>

Limiting initial investment and creating resource scarcity is essential to manage the risk of innovation. Given that most innovative ideas we have will not succeed, we must come up with simple, quick experiments to eliminate bad ideas rapidly and cheaply.

Consider the case of the ARM CPU that is at the heart of almost every mobile device today. The first version of this processor was designed in Cambridge, UK, in the 1980s by two people, Sophie Wilson and Steve Furber. It went from concept to a production-ready design in 18 months.⁸ When their boss, Herman Hauser, was asked how they did it he said, “When we decided to do a micro-processor, in hindsight, I think I made two great decisions. I trusted the team, and gave them two things that Intel and Motorola had never given their people: the first was no money and the second was no people. They had to keep it simple.”

9

The concepts at the heart of the Lean Startup are designed to rapidly evaluate business models through identifying and testing assumptions in a scientific way. Thus they have application beyond the creation of new businesses. For example, the principles of constraining time and resources thus limiting downside, and building a minimum viable product to test your value hypothesis as soon as possible with real customer should be applied at the start of *every* endeavor. We should use this approach to explore *all* new ideas which have unknowns, uncertainties and therefore risks, whether delivering new products, replacing existing systems, adopting new tools and processes, or implementing commercial off-the-shelf software solutions (COTS). Whenever we hear of a new IT project starting up with a large budget, teams of tens or hundreds of people, and a timeline of many months before something actually gets shipped, we have a pretty good idea the project will go over time, budget and not deliver the expected value.

8. http://www.reghardware.com/2012/05/03/unsung_heroes_of_tech_arm_creators_sophie_wilson_and_steve_furber/page2.html

9. <http://www.pcpro.co.uk/features/358750/whatever-happened-to-hermann-hauser>

WARNING

Apply Lean Startup to Internal IT projects

Lean Startup principles are just as important for internal software engineering projects, including internal services and platforms such as private clouds, systems replacements and so forth. We continually encounter enormous initiatives with roadmaps of months or even years spun up to deliver these types of projects, with lip service paid to working incrementally to solve a real (internal) customer problem. In fact, teams building these systems are often dismissive of their customers' needs and preferences — we often hear statements such as “we know better than them what they need”. Projects run in this way, without engaging with their customer throughout the creation process for feedback and collaboration are an appalling waste of time and resources, and rarely achieve their intent, outcome or objectives. But there are other serious negative consequences: having internal systems that are painful to use makes employees frustrated, impacting morale and their ability to do their work effectively. This ultimately costs the business more money through under-performance, and by creating systems that add further complexity to already enormously complex production environments. The inevitable outcome is “shadow IT” — teams deserting services approved or maintained by the IT department in favor of something better so they can get their work done.

Organizations tend to start new projects with large teams both because they assume (wrongly) that this will help finish sooner, and because they use processes, such as annual budgeting cycles, that result in land grabs for favored projects and resources. But when building complex systems these forces inevitably lead to system bloat, increased complexity and dependency management, inefficiency, and poor quality. The cost of establishing and trying to maintain effective communication within large teams consumes enormous amounts of capacity on large projects. Meanwhile, the systems created grow rapidly in an uncontrolled and undirected manner.

In this environment it is extremely hard to establish effective feedback loops to determine whether what is being built is valuable and aligned to the product or project vision. Often it's not even possible to integrate the components to create a working system for much of the project, and when we attempt to do so, we find a myriad of problems that must be addressed to get the system as a whole into a working state, let alone released. It has been our experience, as reflected in [Table 1-2](#), that adding more up-front planning to this process tends to make the eventual outcome worse, not better.

No major piece of work should be fully funded before we have evidence to support the business and economic model on which it is based, and this exploration must be performed using a small, cross-functional team with a limited runway, as described in (to come)

When Exploring New Business Models, Minimize Investment in Software Development.

One large retail organization we worked with wanted to open a store in a new market, their first international expansion. The IT team were given eight weeks to adapt their point of sale system to work in the new country, calculating a different sales tax and using a different currency. We estimated that changing the existing system to work in multiple currencies and calculate taxes correctly would be a substantial multi-month IT project requiring significant investment. Forced to seek viable options to validate the solution was actually possible, the team hard-coded the new sales tax into the existing mainframe system and implemented a simple proxy that replaced the currency symbols in real time for systems in the new store. Although the international expansion was ultimately cancelled as a result of the 2008 financial crisis, the initial software part of the project validated the proposed solution worked with minimal investment, before further investment to create a fully functioning and robust long term solution was agreed.

A final note on exploration. In this chapter we focus on the diffusion of innovation as it applies to products, but exactly the same principles apply to organizational change. Many enterprises try to roll out new methodologies, practices, processes and tools across the whole entire organization in one go, ignoring the fact that people respond to such innovations in different ways, and that there is no “one-size-fits-all” approach to adoption. It is common to see this kind of “big bang” approach fail to achieve the expected results, or be quietly abandoned due to the implementation of the next new initiative which attempts to address the failings of the last.

We should explore and experiment with radical process changes — known as *kaikaku* in Lean terminology — in the same way we explore potential new business models. That is, we should try them out with a relatively small, cross-functional part of the organization, with people that fall in the “innovator” category. These people must be interested in the proposed process experiments,

and have the necessary skills to run them. For changes that prove to be valuable, this team can help other groups adopt these innovations so that they “cross the chasm” within the wider organization until they become the standard way to work. However process improvement does not stop here. All teams will still make continuous, incremental process improvements — known as *kai-zen* — part of their daily work in order to reduce waste and increase throughput and quality, as we discuss in (to come). Organizational culture will be discussed in more detail in (to come).

Exploiting Validated Business Models

Enterprises are optimized to exploit business models that have crossed the chasm — it’s what they are designed to do. However it’s very common for engineering work to be the bottleneck when evolving existing products and introducing new products within the category being exploited.

Projects form the basis of the traditional paradigm for carrying out programs of work in the enterprise. Projects typically require a business cases to be written to gain the allocation of a budget, which in turn leads to a large amount of up-front planning, design, and analysis. The various departments must then must co-ordinate the work and execute the plan. Success of the project is measured by completing the original plan on time and budget. Sadly however, whether the project “succeeds” according to these criteria is irrelevant and insignificant when compared to whether we actually created value for customers, and for our organization.

Data gathered from evolving web-based systems reveals that the plan-based approach does very poorly when attempting to improve valuable outcomes for customers and the organization. Amazon and Microsoft (along with many start-ups) use a technique called A/B testing to gather data on whether a feature will actually deliver value to users before it gets built. Ronny Kohavi, who directed Amazon’s Data Mining and Personalization group before joining Microsoft as General Manager of its Experimentation Platform, reveals the following “humbling statistics”: 60%-90% of ideas *do not improve the metrics they were intended to improve*. Based on experiments at Microsoft, 1/3 of ideas created a statistically significant positive change, 1/3 produced no statistically significant

difference, and 1/3 created a statistically significant negative change.¹⁰ All of the ideas tested were thought to be good ones — but both intuition and expert opinion are extremely poor gauges of the value our ideas deliver to users.

The project paradigm exacerbates this problem. Because projects typically take so long to go from start to finish, stakeholders try and ram as many features as possible into them, mindful of the fact that it will be hard to get features added once the project is complete. Furthermore, the planning process happens when we have the least information and understanding of project risks — right at the beginning. Due to a cognitive bias known as the *planning fallacy*, executives tend to “make decisions based on delusional optimism rather than on a rational weighing of gains, losses, and probabilities. They overestimate benefits and underestimate costs. They spin scenarios of success while overlooking the potential for mistakes and miscalculations. As a result, they pursue initiatives that are unlikely to come in on budget or on time or to deliver the expected returns — or even to be completed.”¹¹

As we execute the project, we discover new information — but since nobody wants their features cut, new information generally leads to more work (known as “scope creep”). Don Reinertsen describes the vicious cycle of adding more scope as we discover more information in the process of running projects as the “large batch death spiral” which, combined with the planning fallacy, means that projects will overrun both their budget and due date in proportion to their size — an important argument for working in small batches.¹²

All of these features and added scope means that projects typically add a tremendous amount of complexity to production environments. This is exacerbated by the fact that “the project phase is often optimized at the expense of the other process areas... and projects are not held accountable for their contribu-

10. (to come)

11. (to come), p252. The planning fallacy is relied upon by many service providers, who submit rock-bottom bids for the initial predefined, contractual services (especially when contracts are awarded to the lowest bidder) and then make their profit through change requests, for which customers pays a premium.

12. Reinertsen devotes a whole chapter ((to come), Chapter 5) of his book to the case for reducing batch sizes.

tions to overall system entropy.”¹³ This complexity leads to higher costs and unplanned work in the operations department, and adds significantly to the cost and effort required to execute future projects.

Finally, because the project approach judges people according to whether work is completed on time and on budget, not based on the value delivered to customers, productivity gets measured based on *output* rather than *outcomes*. This drives several damaging behaviors. Product people become judged on their ability to create comprehensive specification documents and well crafted business cases rather on whether the products and features they come up with deliver value to users. Developers are rewarded for having code completed on their developer workstations, rather than integrated into a working, tested system that can survive real-world usage at scale. We create an unsustainable “hero culture” that rewards overwork and high utilization (making sure everybody is busy) rather than doing the least possible to achieve the desired outcomes. High utilization means work involving collaboration takes longer to complete, because the people you need to work with are always busy with other ‘priorities’. In order to meet ever more serious deadlines, people fail to carry out maintenance and process improvement work (such as automation) that would increase quality and throughput. This, in turn, drives up the cost of doing further work, increasing the pressure on the organization to “work harder”, and fueling a vicious cycle of overwork.

John Seddon, author of *Freedom from Command & Control*, states that “dysfunctional behaviour is ubiquitous and systemic, not because people are wicked, but because the requirement to serve the hierarchy competes with the requirement to serve customers ... people’s ingenuity is engaged in survival not improvement”.

How do we extricate ourselves from this downward spiral? In (to come) of this book, we describe how to run large-scale programs of work in the exploit domain, using the following principles:

1. **Define, measure and manage to outcomes rather than output.** Applying the principle of mission, we specify “true north” for our program of work — our ideal stakeholder outcomes. Then at the program level we work iteratively,

13. (to come), p300.

specifying for each iteration the measurable program-level outcomes we want to achieve. *How* to achieve these outcomes is delegated to teams working within the program. Based on the feedback from real customers after each iteration, we work to improve quality of demand, improve speed, and improve quality of outcomes.

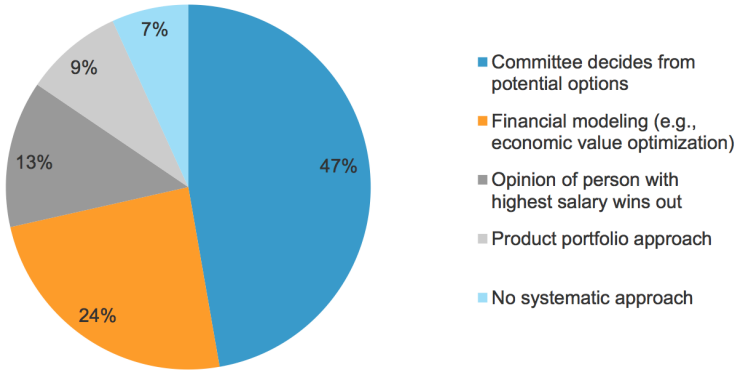
2. **Manage for throughput rather than capacity or utilization.** We implement Kanban principles by making all work visible, limiting work in process, and then aim to stop starting work and start *finishing* it as soon as possible. We continuously undertake process improvement work to reduce lead time — the time it takes to deliver work — through the system. We use continuous delivery to make it cheap and low-risk to deliver work in small increments so that we can deliver work in small batches with increased feedback loops.
3. **Ensure people are rewarded for favoring a long-view system-level perspective over pursuing short-term functional goals.** People should be rewarded for continuous and effective (win-win) collaboration, for *minimizing* the amount of work required to achieve the desired outcomes, and for reducing the complexity of the systems we create to enable these outcomes. People should not be punished when failures occur; rather, we must build a culture of experimentation and collaboration, design systems which make it safe to fail, and put in place processes so we can learn from our mistakes and use this information to make our systems more resilient.

Balancing the Enterprise Portfolio

The key to managing an enterprise business portfolio, as with any financial investment, is to use an economic model. However this is not a widespread practice. In a survey of 161 global business decision makers shown in [Figure 2-6](#), only 24% of respondents reported using an economic model to make investment decisions in products and services. Astonishingly, 13% admitted that the most highly-paid person's opinion (known as the HiPPO method) is the primary deciding factor.¹⁴ 47% reported using the only slightly less embarrassing method of decision-by-committee.

14. This term was coined by Ronny Kohavi, partner architect at Microsoft.

“Please select the statement that most closely aligns with how your company decides which products are built?”



Base: 161 business decision-makers

Source: A commissioned study conducted by Forrester Consulting on behalf of Thoughtworks, September 2012

Figure 2-6. *How do enterprises make investment decisions?*

In *Escape Velocity: Free Your Company's Future from the Pull of the Past*, Geoffrey Moore presents a “growth / materiality matrix” for visualizing existing investment decisions, shown in [Figure 2-7](#), and describes how it allows us to distinguish between companies which have an effective portfolio strategy and those which do not. The y axis of the diagram measures whether a particular business is material to your business relative to others, where “material” means that it generates 5-10% or more of total revenue or profit. The x axis measures the growth rate of the business in absolute terms.

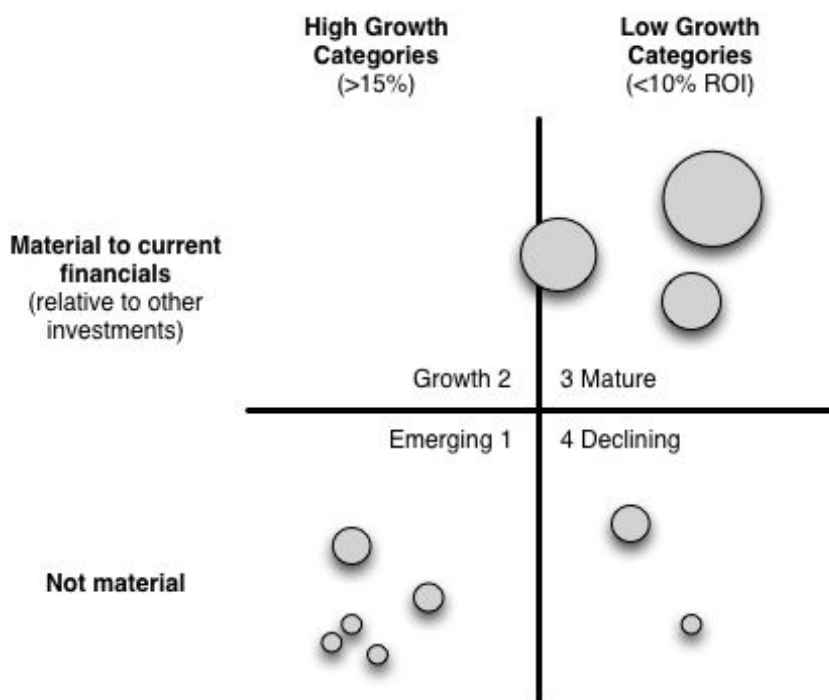


Figure 2-7. *The Growth / Materiality Matrix for Portfolio Management*

Many organizations — in 2014, think Microsoft, IBM, and HP — have market leading franchises in quadrant 3, corresponding to stage C (a mature market) in [Figure 2-3](#). But, as shown in [Figure 2-7](#), none has been able to develop (rather than acquire) a major franchise in stage B (corresponding to quadrant 2) despite substantial R&D investments which have led to new businesses in quadrant 1. In contrast, Amazon, Google and Apple have each grown businesses in the decade to 2014 which have grown rapidly to become significantly material to the enterprise.

In order to understand why so many companies fail to create businesses in quadrant 2, we must understand the dynamics of the enterprise portfolio. This is described in the *three horizon* model presented in (to come), shown in [Figure 2-8](#). Horizon 1 consists of your set of core product categories and businesses (corresponding to quadrant 3 above). Investments in horizon 1 businesses will deliver results in the same year, and typically take the form of developing existing

products and launching new ones within the existing categories. Horizon 2 is the set of emerging businesses which will form the core business of the future. These require significant investment and require the attention of sales and marketing divisions to succeed, but will not deliver the same levels of returns as Horizon 1 investments.

Horizon 3 is the domain of the Lean Startup, where we experiment with new business models and attempt to create product/market fit for new businesses. We aim to invest just enough time and money to get sufficient information to make an investment decision. We then either move the idea into Horizon 2 or shelve it, perhaps until market conditions or advances in technology make it favourable to try again.

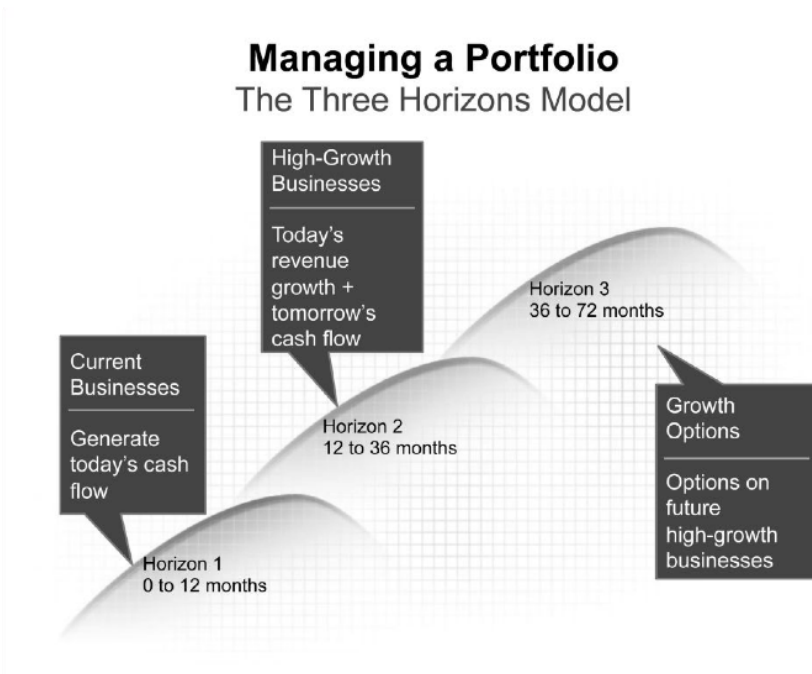


Figure 2-8. *Three Horizons*

There are three significant problems which conspire to kill businesses that make it through horizon 2. First, they require substantial investment in terms of research, sales and marketing resources without delivering corresponding revenue returns—the metric by which these departments are usually measured.

Second, each of the three horizons requires very different management practices and support in order to succeed, as shown in Table 2-2. Blindly applying a consistent approach to each will result in failure. Finally, as Clayton Christensen describes in *The Innovator's Dilemma*, profitable enterprises are often reluctant to cannibalize their profits and market share by launching disruptive new products — defined as those which might threaten their existing bottom line and market valuation.

Table 2-2. *Three Horizons*

	Horizon 1 (0-12 months)	Horizon 2 (12-36 months)	Horizon 3 (36-72 months)
Goals	Maximize economic returns	Cross the chasm, start contributing significant revenues	Create a new business
Key metrics	Revenue vs plan, market share, profitability	Rate of sales, target accounts	Buzz / word-of-mouth popularity (consumer), Name-brand customers (enterprise)
Metrics	Growing category, Share, Net promoter, Revenue	Growth, Increasing efficiency (will lead to profitability)	Love Metrics based on delivering the customer benefit, Active product usage, Proactive word of mouth

We frequently see these forces conspiring to prevent businesses making it through horizon 2, even in enterprises which do an excellent job of both exploring and exploiting business models. Often, other companies eventually bringing them to market with devastating results. Xerox PARC invented the modern GUI (as well as many other elements of modern computing), but the “toner heads” at Xerox’s head office “had no clue about a computer or what it could do” and so it was ultimately Apple and Microsoft who brought computers into people’s living rooms instead.¹⁵

Photography giant Kodak, which filed for bankruptcy in 2012, actually invented the digital camera. Steve Sasson and his team from the Kodak Apparatus Division Research Laboratory created the breakthrough innovation in 1975.¹⁶ However the team was met with puzzled managers who could not comprehend

15. Steve Jobs, *The Lost Interview*

16. <http://www.techradar.com/news/cameras/photography-video-capture/how-kodak-invented-the-digital-camera-in-1975-364822>

the future opportunity or why customers would ever want to view photographs on a monitor. Their business was optimised for developing photographs — producing paper, film and other supplies — not capturing memories.

WARNING

Why You Cannot Simply Hire or Acquire Your Way To Innovation

To pick on a current trend, a number of enterprises have been acquiring startups in an attempt to accelerate innovation, perhaps by turning them into “innovation labs” or tick a box in stage A, quadrant 1 or horizon 3 to diversify and balance their portfolio. We have seen the poor outcomes this creates at close hand, with these acquisitions failing to produce the expected return, and the senior staff leaving as soon as they could exercise their options. The problems occur when the acquired company — working on a horizon 3 or 2 product — is subjected to the horizon 1 governance, financial target, and management structures of the acquiring enterprise, thus completely destroying its ability to innovate. Sometimes people from the parent organization are rotated through the innovation lab in the hope that this will magically teach them how to innovate in the context of a different horizon, rather than simply giving them culture shock. Acqui-hiring frequently fails for the same reason: taking great people and putting them into a pathological or bureaucratic culture does not change the culture, it breaks the people. The solution is to do the hard work to transform the culture of your own organization and create a vision which motivates employees and customers — which will, incidentally, remove the need to hire in or acquire a capability to innovate.

Our hypothesis is that organizations survive and grow in the medium and long term by balancing the ability to continuously explore potential new business models with effective exploitation of existing ones. Indeed, one of the hallmarks of a truly adaptive and resilient organization is that it continually disrupts its own existing business models in search of future opportunities and new markets and customers.

For example, Amazon’s pursuit of electronic books and the production of the Kindle disrupted what was then its primary business model of selling physical books. The development of the Amazon Marketplace enabled other vendors to leverage Amazon’s infrastructure to potentially undercut products sold by Amazon. 3M defines its strategy as constant new product innovation and sets targets for the percentage of revenue from products introduced in the past five years

to be 30%, which it exceeded in 2008.¹⁷ Inge G. Thulin, President and CEO of 3M, expects that number to reach 40 percent by 2017.¹⁸

Intuit uses a simple model to balance horizons 1, 2 and 3, as shown in [Table 2-3](#). Google follow a similar model, but with different allocations: 70% to horizon 1, 20% to horizon 2, and 10% to horizon 3.¹⁹

Table 2-3. *Intuit Innovation Horizons and Metrics*

	Existing businesses	Adolescent businesses	Ideas
Investment	60%	30%	10% of operating expenses, funded quarterly based on validated learning
Metrics	Growing category, Share, Net promoter, Revenue	Growth, Increasing efficiency (will lead to profitability)	Love Metrics based on delivering customer benefit, active product usage, proactive word of mouth
Example products	TurboTax, Mint	QuickBooks Online Accounting	SnapTax

20

The most important point to bear in mind when balancing horizons is that unless senior leadership takes an active role in managing investments, including putting in place appropriate management practices for different horizons and paying attention to how senior management is incentivized, core businesses will always find a way to use their superior corporate clout to sideline and ultimately neutralize the other horizons. If the cultural and management barriers are simply too strong for this kind of “ambidextrous” approach to succeed, the alternative is to spin off a maximally independent business unit.²¹

17. http://money.cnn.com/2010/09/23/news/companies/3m_innovation_revival.fortune/

18. <http://www.cnbc.com/id/100801531>

19. http://money.cnn.com/magazines/business2/business2_archive/2005/12/01/8364616/index.htm

20. <http://www.leapfrogging.com/wp-content/uploads/2013/05/Intuit-Innovation-Catalyst.pdf>

21. For more on “ambidextrous organizations”, see (to come).

How Aetna Created New Companies To Disrupt its Core Businesses

Aetna, like all the players in the US healthcare market, knew that the Obama administration's Affordable Care Act represented both a serious risk and a significant opportunity. 160 years old at the time the bill was signed into law, Aetna decided to create a new company called Healthgen, "a separate organization, separately capitalized, separately compensated, and separately managed, so they're not subject to the same management process at Aetna" with the purpose of disrupting the healthcare provider market with new technology and business models. Healthgen has a goal to drive \$1.5bn - \$2bn of revenue per year initially.

22

Aetna has also created another subsidiary along similar lines to create a consumer marketplace and drive private exchange models. Mark Bertolini, Aetna's Chairman, President and CEO, states that his goal is to build a technology-based competitive ecosystem that will disrupt Aetna's own core business.

Conclusion

In this chapter we started by introducing the lifecycle of business models, technologies, and ideas. We then described the very different behaviours required to explore new technologies and ideas compared with exploiting proven ones. Finally, we described the importance of balancing your investment across these domains in addition to investments in your core businesses, and discussed how to apply the three horizon framework to balance these investments.

We identified how organizations can effectively manage their portfolio of investments:

- Use an economic model to balance investments across the three horizons, and set targets for each horizon.
- Recognize that different horizons require different management practices, and ensure that they are put in place.

22. <http://teconomy.com/conf/13-tucson/business-revolutions/conversation-mark-bertolini/>

- Use the Lean Startup method in combination with the principle of optionality to discover new businesses
- Exercise the following principles when exploiting validated businesses: define, measure and manage to outcomes rather than output; manage for throughput rather than capacity or utilization; ensure people are rewarded for favoring a long-view system-level perspective over pursuing short-term functional goals.
- Begin *any* major investment with an “explore” phase which involves solving a real problem for customers and demonstrating product/market fit (even when the market and customers are internal to your enterprise).

However it’s essential to remember that enterprises will not be able to successfully adopt these principles unless leaders first grow an innovation culture as described in [Chapter 1](#).

For further reading on portfolio management, we recommend Geoffrey Moore’s *Escape Velocity: Free Your Company’s Future from the Pull of the Past*.

Questions for readers:

- What framework does your organization use to balance your portfolio of exploring new business models, exploiting validated ones, and developing core businesses?
- Is there somewhere where you can see this portfolio at a glance?
- What performance metrics are used to measure the health of activities in each of these domains?
- What is the percentage ratio for investments in horizon 1, 2 and 3 in your organization? Is it intentional or accidental? What do you think should it be?
- How does senior leadership ensure that investments in horizons 2 and 3 are nurtured, and that transitions between horizons are managed in a way that will maximize the relevant performance metrics for each individual investment?

Experimental By Design

Doubt is not a pleasant condition, but certainty is absurd.

— VOLTAIRE

For enterprises experimenting with new business models and products, as for startups, the biggest risk is a failure to create something that actually delivers value to users. The Lean Startup provides a framework that allows us to rapidly discard ideas that do not deliver value or will not be adopted sufficiently quickly so we don't waste our resources on them. However the principles behind the Lean Startup can be applied to *all* kinds of activities within the enterprise, such as building internal tools, process improvement, organizational change, systems replacement, and GRC programs.

In this chapter we present the principles and concepts that enable us to take a systematic approach to managing the risk of planned work, by gathering information to reduce uncertainty. This forms the basis of the practical approach we present to exploring new opportunities we present in the rest of (to come).

Model Investment Risk

Typically in enterprises we must build a business case along with a plan to support it before we can get approval to proceed. This usually involves a team of people creating a detailed document that estimates the value the proposed initiative will create. The business case describes the required resources and dependencies, and finally a beautifully crafted set of numbers detailing the planned

work with costs, key metrics, dependencies, a resource plan, and timeframes. Depending on the level of detail and the estimated investment required, this process can take weeks or months to complete.

An important goal of the planning process is to support an investment decision. In order to make this decision, we need to have a good understanding of the *risks* involved with the investment. Following Douglas Hubbard, we define risk as “A state of uncertainty where some of the possibilities involve a loss, catastrophe, or other undesirable outcome,” and the *measurement* of risk as “A set of possibilities, each with quantified probabilities and quantified losses,”¹ for example, “we believe there is a 50% chance the project will be cancelled, with a loss of \$2m in development work.”

In *How to Measure Anything*, Hubbard discusses his work analyzing business cases for IT investments:

Each of these business cases had 40 to 80 variables, such as initial development costs, adoption rate, productivity improvement, revenue growth, and so on. For each of these cases, I ran a macro in Excel that computed the information value for each variable. I used this value to figure out where to focus measurement efforts. When I ran the macro that computed the value of information for each of these variables, I began to see this pattern: 1) The vast majority of variables had an information value of zero... 2) the variables that had high information values were routinely those that the client never measured. 3) The variables that clients used to spend the most time measuring were usually those with a very low ... information value.

2

Take the example of estimating development costs in order to put together business cases to obtain project approval. This usually involves analyzing months worth of future work, breaking it into small pieces and estimating the effort required for each piece. However, as Hubbard notes, “Even in projects with very uncertain development costs, we haven’t found that those costs have a significant information value for the investment decision... The single most important unknown is whether the project will be canceled... The next most impor-

1. Definitions are taken from (to come), p50.

2. (to come), p111

tant variable is utilization of the system, including how quickly the system rolls out and whether some people will use it at all...”³

Thus the business case essentially becomes a science fiction novel based in an universe that is poorly understood — or which may not even exist! Meanwhile significant time is wasted performing detailed planning, analysis and estimation, which provides large amounts of information with extremely limited value. According to research by Donald Reinertsen, author of *The Principles of Product Development Flow: Second Generation Lean Product Development*, it's typical for 50% of total product development time to be spent in such “fuzzy front end” activities. Needless to say, this leads to poor investment decisions and needlessly long product development cycle times. This creates multiple negative outcomes:

- Typical market research activities are poor at predicting product / market fit, especially in new product categories. Research said that minivans and iPods would not be successful.
- In the absence of good data, people tend to get their pet projects funded. Particularly in enterprise IT, we often see spectacular amounts of money poured down the drain on systems replacement projects — even (perhaps especially?) in organizations operating in highly regulated sectors.
- Long product development cycle times dramatically reduce the potential return on investment we can achieve from successful new products.
- Most perniciously, long product development cycles delay the time it takes to get customer feedback on whether we are building something valuable.

There are two factors we care about in a business plan. The first is the sensitivity of the key metric (for example lifecycle profit in the case of a new product) to the various variables in the business case. The second is the level of uncertainty in the variables to which the key metric is sensitive. Given distributions and ranges for the key variables, a simple but powerful approach is to perform a Monte Carlo simulation to work out the possible outcomes. This then allows us to find the variables to which we need to pay further attention in order to make good investment decisions.

3. http://www.cio.com/article/119059/The_IT_Measurement_Inversion

To run a Monte Carlo simulation, we use a computer to create thousands of randomized possible scenarios based on the distribution shape and ranges for the input variables, and then compute the value of the metric we are interested in for each scenario. The output of a Monte Carlo simulation is a histogram, with the number of scenarios for each range on the y-axis, and the ranges on the x-axis. You can perform a Monte Carlo simulation using Excel, or use one of a number of the custom tools that exist.⁴ The output of a Monte Carlo simulation for a business case might look something like [Figure 3-1](#). As Hubbard notes, the uncertainty in ROI for IT programs tends to be very high, and increases in line with the duration of the program.

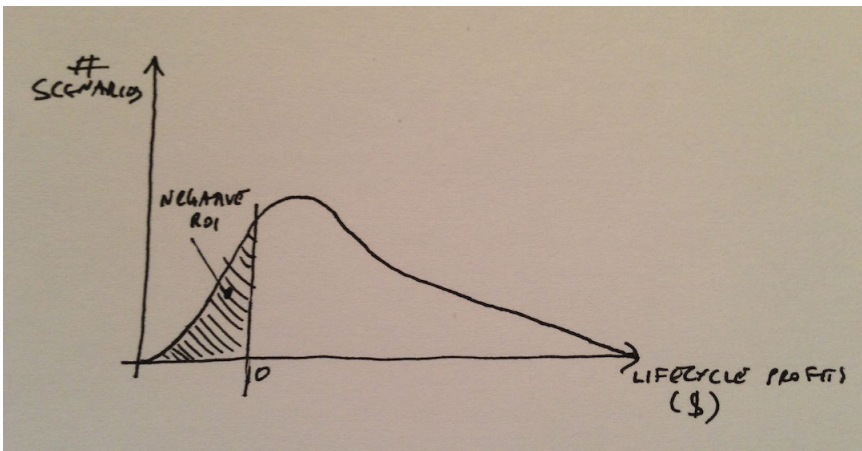


Figure 3-1. Output of a Monte Carlo simulation

As Hubbard notes, and as you can verify by doing a Monte Carlo simulation on your own business cases, ROI in IT programs is not very sensitive to cost, but rather whether the program will be cancelled, and the utilization of the resulting system. These variables depend primarily on whether we have *built the right thing*. However the standard enterprise planning process provides *almost no validation of this*.

Let us be absolutely clear. *In most enterprises, around 30%-50% of the total time to market is spent on activity which provides almost zero value in terms of*

4. see <http://www.howtomeasureanything.com/> for an example.

mitigating the risk of our investments. This near-zero value add activity is mostly driven by financial management and planning processes. In our experience, the fuzzy front end presents the biggest opportunity for radical process improvement (*kaikaku*) in enterprises. We can drastically reduce the amount of time required, and make better decisions, by taking a systematic approach to risk management. In this chapter, we discuss how to attack the fuzzy front end for new businesses and new products. In (to come) we show how to change the way program-level feature backlogs are managed.

Applying the Scientific Method to Product Development

The way the world tells you whether what you are doing is valuable is whether they send you money.

— DONALD REINERTSEN

When there is a large amount of uncertainty in the key metric we care about, we begin by identifying the variables with the highest information value, the riskiest assumptions. These are the ones to which our outcome metric is most sensitive. In the case of both business model innovation and product development, Donald Reinertsen comments that in the context of product development, “unit sales are where the bodies are buried.”

The most inefficient way to test a business model or product idea is to plan and build a complete product in order to discover whether the predicted market for it really exists. Yet this is exactly what we do once we have an approved business case. Part of the problem is the language we use to describe the product development process. For example consider the term “requirements.” Whose requirements are they? Are they the user’s requirements? In *Lean IT*, Steve Bell and Mike Orzen comment that “Users are often unable to articulate exactly what they need, yet they often seem insistent about what they don’t want ... once they see it.”⁵

We should stop using the word “requirements” in product development, at least in the context of non-trivial features. What we have, rather, are *hypotheses*. We *believe* that a particular business model, or product, or feature, will prove val-

5. (to come) p48.

uable to customers. But we must test our assumptions. We can take a scientific approach to testing these assumptions by running experiments.

In the case of business model and product innovation, the Lean Startup movement provides us with a framework for operating in conditions of extreme uncertainty. In *Running Lean*, Ash Maurya presents how to execute the Lean Startup model:

- Rather than spending a lot of time creating a sophisticated business model, we design a simplified *business model canvas* which captures and communicates the key operating assumptions of our proposed business model.
- We gather information to determine if we have a problem worth solving — meaning that it is both solvable, and people will pay for it to be solved. If both of these conditions obtain, we have achieved *problem/solution fit*.
- We then design a *minimum viable product* (MVP) — an experiment designed to maximize learning from potential early adopters with minimum effort. In the likely case that the results of the MVP invalidate our product hypothesis, we *pivot* and start again. We continue this process until we decide to abandon the initial problem, run out of resources, or discover *product/market fit*. In the latter case we exit the explore phase and proceed to exploit the validated model.
- Throughout this process, we update the business model canvas based on what we learn from talking to customers and running MVPs.

We present this approach in detail in [Chapter 4](#).

There are two key innovations in this model. First of all, we stop using detailed planning as a way to manage risk. Instead we find customers and run cheap experiments designed to discover if our proposed business model or product is actually valuable to them. Second, rather than creating only one plan, we iterate by running a *series* of experiments in order to discover product/market fit, since we expect that in conditions of uncertainty our first idea is very unlikely to bear fruit.

A common objection to these principles is that such experiments cannot possibly be representative of a complete product. This objection is based on a false understanding of measurement. The purpose of measurement is not to gain certainty, but to *reduce uncertainty*. The job of an experiment is to gather *observa-*

tions that quantitatively reduce uncertainty.⁶ The key principle to bear in mind is this: *where the level of uncertainty of some variable is high, we need very little information to significantly reduce our uncertainty.*

NOTE

Definition of Measurement

Measurement: A quantitatively expressed reduction of uncertainty based on one or more observations.⁷

This definition may seem counterintuitive unless you have experience running experiments in a scientific context. In experimental science, the result of a measurement is never simply a single value. It is, rather, a *probability distribution* which represents the range of possible values, as shown in Figure 3-2. Any measurement that doesn't indicate the precision of the results is considered practically meaningless. For example, a measurement of my position with a precision of 1 meter is far more valuable than that same position with a precision of 500 miles. The point of investing in measurement in a scientific context is to *reduce our uncertainty* about the actual value of some quantity. Thus, in particular, if we attempt to state estimates as a precise number (as opposed to a range), we are setting ourselves up for failure: the chance of us meeting a date 6 months in the future *precisely to the day* is practically zero.

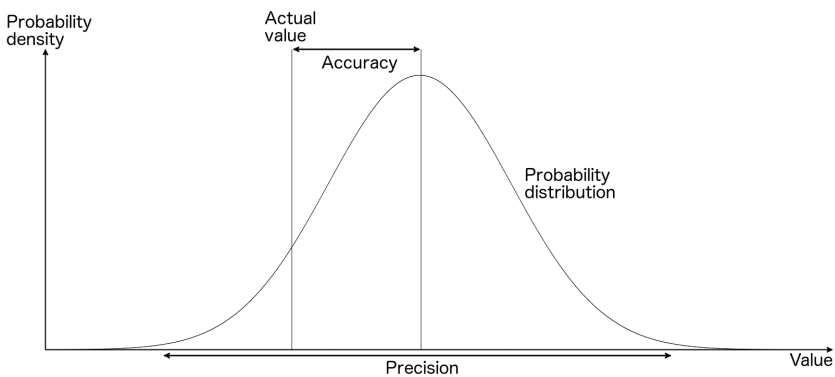


Figure 3-2. Accuracy and precision

6. (to come) p23.

7. (to come) idem.

A minimum viable product can be thought of as a way to conduct a relatively cheap measurement so as to reduce our uncertainty concerning our key metric. This is what makes an MVP such a good investment. Typically putting together a business plan and requirements for a significant investment takes weeks or months in an enterprise context. In the same amount of time, by following the Lean Startup model, we could be running multiple experiments, learning from real customers and emerge with a superior, battle-tested plan based on evidence. Let’s examine the differences between these two approaches at the point we need to make the investment decision:

Table 3-1. *Traditional product lifecycle vs Lean Startup lifecycle*

	Traditional project planning process	Lean Startup discovery process
<i>What data do we have to make the investment decision?</i>	A business plan based on a set of untested hypotheses and assumptions, backed by case studies and market research.	Real data based on evidence compiled from a working product or service tested with real customers.
<i>What happens next?</i>	We must create detailed requirements, if we haven’t already, and then start a project to build, integrate, test and finally release the system.	We already have a validated MVP which we can build upon immediately with new features and enhancements based on customer feedback.
<i>When do we find out if the idea is any good (i.e. will it get a good return on investment)?</i>	Once the project is complete and the product or service is live.	We already have this evidence based on the data we have collected.

As discussed in [Chapter 2](#) an important factor in the success of the Lean Start-up approach is to limit the size of the explore team and the resources available to them (including time). This encourages people to apply their creativity and focus on learning rather than the pursuit of “perfect” solutions. There are no awards for elegance of software design or automated test coverage in an MVP — the more skeletal, the better, provided we can gather the information we need. Many of the “war stories” exchanged by Lean Startup practitioners describe the ingenious shortcuts they took in the pursuit of validated learning.

Of course a reasonable question is: Given that product development is effectively a form of discovery, how much time and money should we spend on validated learning? Game theory actually provides a formula for the *expected value of information* (EVI). A detailed discussion of how to calculate this number is beyond the scope of this book, but it is covered in Hubbard’s *How to Measure*

Anything ⁸. The EVI gives us an upper bound on how much we should be prepared to pay to gather the information in question. If the cost of performing a measurement to gather the information is much less than the EVI (say, an order of magnitude), it is clearly worth performing the measurement. Thus the more risky and expensive the project in question, the better the value for money provided by pursuing a Lean Startup approach.

NOTE

Expected Value of Information

Hubbard defines the value of information as follows: “Roughly put, the value of information equals the chance of being wrong times the cost of being wrong. The cost of being wrong — that is, what is lost if your decision doesn’t work out — is called an opportunity loss. For a simplistic example, say you’re considering investing \$1 million in a new system. It promises a net \$3 million gain over three years. (For our example’s sake, it’ll either be completely successful or a total bomb.) If you invest but the system fails, your mistake costs you \$1 million. If you decide not to invest and you should have, the mistake costs you \$3 million. When we multiply the opportunity loss by the chance of a loss, we get the expected opportunity loss (EOL). Calculating the value of information boils down to determining how much it will reduce EOL.” ⁹

In reality, the success of a product is rarely a binary outcome. If we return to the example of the predicted ROI for a business case illustrated in [Figure 3-1](#), we can calculate the EOL by calculating the area of the shaded part of the curve, which represents the scenarios in which we lose money on our investment. Thus we sum up the ROI at each point multiplied by the probability of that outcome. Assuming we had perfect information which predicted the exact outcome in ROI, that could potentially be worth as much as the EOL we have just calculated. Since an MVP will typically provide less than perfect information, the EOL represents an upper bound on what we should spend on the runway for discovering product/market fit. ¹⁰

8. (to come), Chapter 7

9. <http://www.cio.com/article/2438748/it-organization/the-it-measurement-inversion.html>

10. Hubbard provides a spreadsheet which enables you to calculate the value of information on his website, <http://howtomeasureanything.com/>

Applying the Lean Startup approach internally within enterprises

The Lean Startup model isn't limited to new product development: it can be used for any kind of new work in an enterprise context, including systems replacement, building internal tools and products, process innovation, and evaluating commercial off-the-shelf software (COTS). In all cases we begin by stating our goal in terms of the *measurable customer outcome* we wish to achieve. Ideally this will be in terms of end users and customers outside the enterprise, but we can define it in terms of our immediate downstream customer. For example, for an internal test automation tool, we might aim to reduce lead time for full regression testing to 8 hours.

To determine if we have problem/solution fit, we look for a customer who will be willing to work with us to pilot the new system, tool, process or software. This is a critical step which is often skipped by enterprises. Indeed for internal tools it's common to *mandate* use of them, a disastrous policy which often results in enormous amounts of waste, unhappy users, and little value to the organization. The process of finding a customer and discovering a real problem that they will pay you to solve (even if the payment takes the form of time and feedback rather than money) — and thus obtaining problem/solution fit — is essential to developing internal tools, purchasing COTS, and internal systems replacement. Mandating the use of a particular solution makes it much harder to gather the essential feedback on whether that solution is actually providing value.

Once we have a pilot team, we then design and execute a minimum viable product. This may be a prototype of a tool designed to help just one team, or an implementation of a COTS package that solves a problem for just one team, or for a single business process for that team. The hardest part here is to limit scope so as to solve a real problem but deliver something in the space of days or weeks, rather than months. The worst thing we can do is disappear to design the perfect tool or adoption strategy, without continually delivering value to real users and gathering feedback from them throughout the process. It's essential to be disciplined about time-boxing this activity, and to focus on solving a real and an urgent problem as soon as possible.

The measure of success — and whether or not we should proceed — is whether our users find the MVP good enough to use of their own free will, and whether we

actually meet the measurable customer outcome we set out to achieve. If not, we need to pivot and return to the beginning.

Principles for Exploration

In [Chapter 1](#) we showed how small, highly motivated forces were able to defeat larger, better trained enemies through a style of war known as maneuver warfare. “Disruption” is a word that is currently ubiquitous to the point of cliché, but in the context of maneuver warfare the chief exponent of the idea of disrupting your opponent’s decision-making process was Colonel John Boyd of the US Air Force. In his career as a fighter pilot and instructor, Boyd was famous for never losing his bet that he could win any dogfight—from a position of disadvantage—within 40 seconds, and also for co-creating the energy-maneuverability theory of aircraft performance that led to the design of the F-16 fighter jet. However his best-known creation is the “OODA loop,” shown in [Figure 3-3](#), a model of how humans interact with their environment which forms the basis of Boyd’s theory of maneuver warfare. OODA stands for *observe*, *orient*, *decide*, *act*, the four activities that comprise the loop.

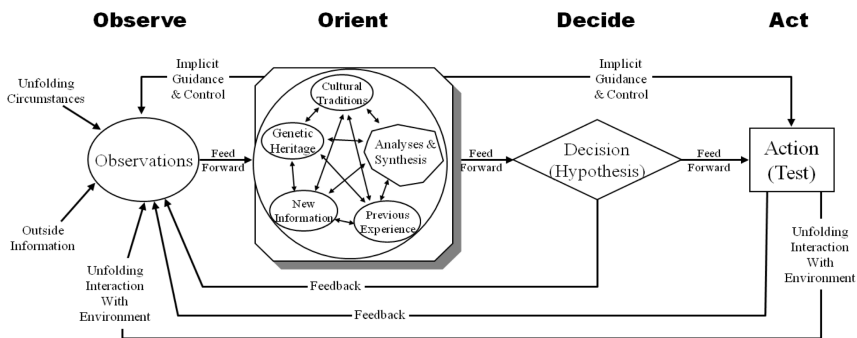


Figure 3-3. The OODA loop

A common misconception (made primarily by people who have not actually seen the diagram) is that these activities are carried out one after the other in a loop, and that disruption is achieved by going through the cycle faster than your opponent. There are two important flaws with this interpretation. First, in reality both humans and organizations are performing all of these activities simultaneously, and there are multiple feedback and feed-forward loops between each of

them. Second, it is often advantageous to *delay* making decisions until the “last responsible moment” (which we can analyze using optionality and Cost of Delay, for which see (to come)).

To truly understand the diagram, we must start with *orientation*. Boyd’s insight here is that our observations, decisions, and actions are all contingent upon our current orientation, which is in turn determined by a complex series of factors including our genetics, our habits and experiences, and the cultures within which we grew up and are currently operating, as well as the information we have to hand. The second thing to note about the diagram is that there are two mechanisms of influence: one is the feedback and feed-forward loops, and the other is “implicit guidance and control.”

Psychology tells us that our actions can be shaped either by IGT (implicit guidance and control), or by feed-forward from a conscious decision. Implicit guidance and control in humans is provided by a system in the mind called *System 1*, which “operates automatically and quickly, with little or no effort and no sense of voluntary control.” Conscious decisions are made by *System 2*, which “allocates attention to the effortful mental activities that demand it, including complex computations. The operations of System 2 are often associated with the subjective experience of agency, choice, and concentration.”¹¹ Equally, IGT affects how we *observe* things, for example our tendency to ignore information which contradicts our beliefs (a cognitive bias known as *confirmation bias*).

Both of these mechanisms exist at the organizational level. In terms of action, organizations use the implicit guidance and control mechanism when we delegate decision-making using decentralized command and the principle of mission, relying on a shared understanding of our goals along with alignment across the organization to ensure that people act in the interests of the wider organization. However some actions (particularly ones involving compliance) must be taken using the explicit feed-forward mechanism.

Implicit guidance and control also governs how organizations observe. Generative cultures create monitoring systems and visible displays that enable people throughout the organization to rapidly access relevant information which, in turn, changes their orientation. Changes in orientation will cause us to update what we measure and how information flows through the organization. In

11. (to come), pp20-21. These names were coined by Stanovich and West in (to come).

pathological and bureaucratic organizational cultures, measurement is used as a form of control, and people hide information that challenges existing rules, strategies, and power-structures. As Deming said, “whenever there is fear, you get the wrong numbers.”

When Boyd talks about “operating inside” an opponent’s OODA loop, he means *understanding* our opponent’s loop and how it will determine their actions, and then using that knowledge against them:

The basic pattern is simple: An organization uses its better understanding of — clearer awareness of — the unfolding situation to set up its opponent by employing actions that fit with the opponent’s expectations, which Boyd, following Sun Tzu, called the zheng. When the organization senses (viz. from its previous experiences, including training) that the time is ripe, it springs the qi, the unexpected, extremely rapidly. The primary reason for implicit guidance when engaged with opponents is that explicit instructions — written orders, for example — would take too much time. As Boyd put it, ‘The key idea is to emphasize implicit over explicit in order to gain a favorable mismatch in friction and time (i.e. ours lower than any adversary’s) for superiority in shaping and adapting to circumstances.’¹²

The OODA model can also be applied also in the context of customer engagement: “Instead of surprise → shock → exploitation, as in war and the martial arts, zheng/qi could operate as something more like surprise → delight → fascination → become more committed customers. Apple plays this game, the ‘pursuit of wow!’ as Tom Peters once described it, very well.”¹³

Boyd refers to the implicit guidance and control pathways within an organization, determined by its culture and existing institutional knowledge and processes, as its *repertoire*. We have discussed how organizations apply their existing repertoire to disrupt competitors, but in order to improve performance and avoid disruption, we must constantly be creating new repertoire of our own. This can

12. This quote, along with the OODA loop diagrams in this section, are taken from Chet Richards’ excellent discussion of the OODA loop: http://www.jvminc.com/boydsrealooda_loop.pdf. Chinese words have been updated to use pinyin.

13. Chet Richards, op. cit.

take the form of process improvement, training, the evolution of existing products, and the creation of new businesses and new products. This loop is also represented in the OODA model, as shown in [Figure 3-1](#).

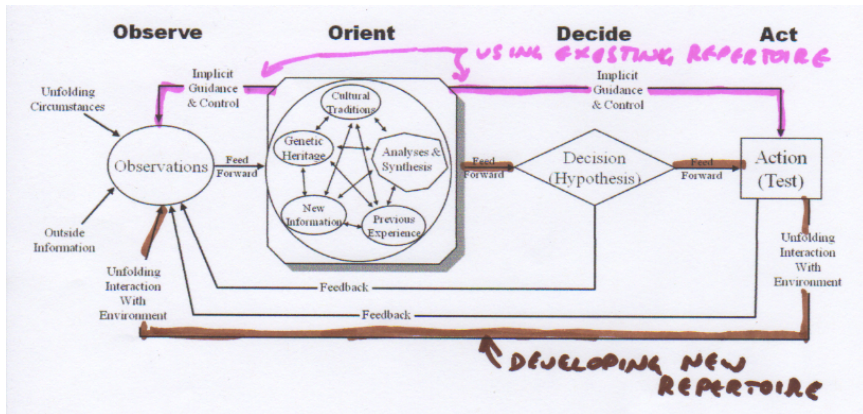


Figure 3-4. *Creating new repertoire*

The repertoire generation loop is more or less a statement of the scientific method, in which we create new hypotheses based on observation and synthesis, design experiments to test these hypotheses, and then update or discard theories (which form part of our orientation) based on the results of the experiment. This loop, in turn, inspired Eric Ries' Build-Measure-Learn loop, shown in [Figure 2-4](#), which shows us how to create new repertoire in the form of new business models, products, and features. The build, measure, learn loop seems straightforward, but is hard to adopt in practice due to its combination of a scientific approach (building to learn) with an engineering mindset (learning to build).

When used for process improvement (discussed in (to come)) and for changing organizational culture (discussed in (to come)), we can use a loop known as the Deming cycle, shown in [Figure 3-5](#).

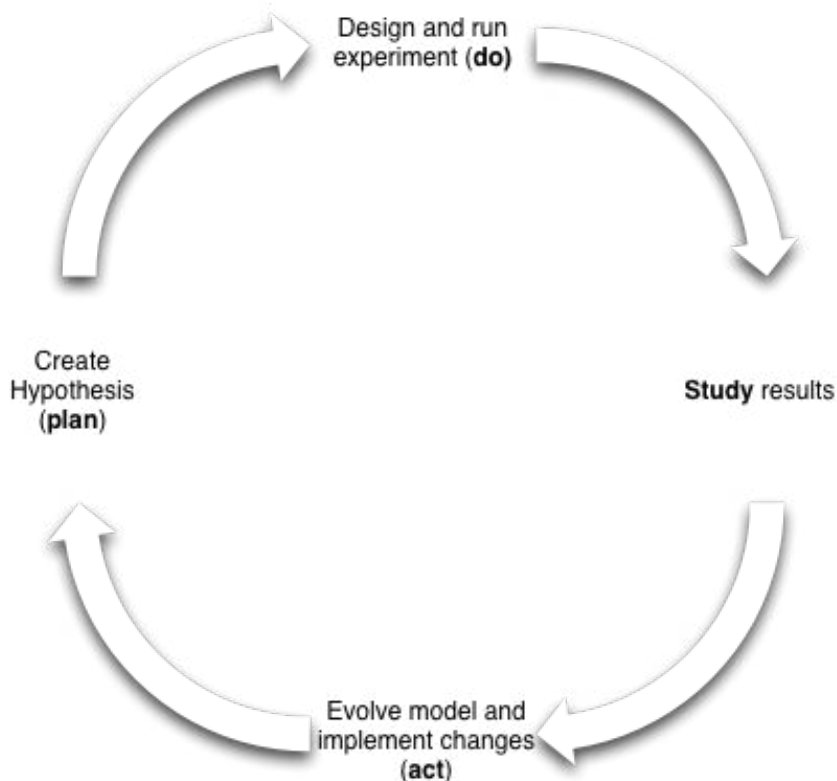


Figure 3-5. *The Deming Cycle*

The key to being successful with these cycles (and the scientific method generally) is to use them *systematically* and *continuously*. Applying them *systematically* means we use them as a general tool to explore *all* types of risk (ensuring that the expense of running the experiment is commensurate with the value of the information we will discover). Applying them *continuously* means as often as possible (as Mike Roberts says, “continuous means much more often than you think”), with a focus on getting through the loop in the shortest possible time. The most important question to ask in the context of repertoire generation is: how fast can we learn? While we may not immediately *release* the results of our learning exercises to the wider world — when to launch your product is a matter of strategy — we should constantly be working to learn and test our assumptions with real users as frequently as possible.

When everybody in the organization has been trained to employ the scientific method to innovate as part of their daily work, we will have created a generative culture that allows the organization to adapt rapidly to its changing environment. Toyota calls this “building people before building cars.”¹⁴ It’s essential to distinguish between Taylor’s Scientific Management, as discussed in [Chapter 1](#), and the experimental approach we discuss here. In Scientific Management, analysis is performed and decisions taken by management, with the people doing the work functioning more or less as automatons. In the experimental approach we describe in this book, the job of leadership and management is to design, evolve and operate a system in which the people doing the work have the necessary skills and resources to run their own experiments, thus individually and collectively learning, developing, and growing their knowledge by performing the experiments themselves.

As shown in [Table 3-2](#), applying the scientific method to product development is a fundamentally different approach to the traditional plan-based approach and requires different skills and behaviors. It is not that the traditional project lifecycle is bad — it can be effective when used in the context of projects where the thing to be built has been built many times before and the risks are well understood. But the tradition is the wrong model for working in conditions of uncertainty, such as product development or any kind of custom software development.

Table 3-2. *Traditional project lifecycle vs Lean Startup lifecycle*

Skill or behavior	Traditional planning approach	Experimental approach
<i>Attitude to plan</i>	Changes to the plan once it has been agreed upon are considered problematic and indicate a failure in the process	We expect that the initial plan will not survive contact with real customers, and aim to invalidate it and pivot as quickly as possible
<i>Skills required</i>	Requirements gathering, analysis, determination of costs, resource and dependency planning, ability to gain political support	Designing experiments and performing measurements, data collection and analysis, ability to work effectively in cross-functional teams and communicate with wider organization

14. (to come).

Skill or behavior	Traditional planning approach	Experimental approach
<i>How success is measured</i>	Whether plan is approved and funded	How fast we can go through learning cycles and exit the explore phase, either by deprioritizing or cancelling the work, or proceeding to the exploit phase
<i>How we achieve compliance</i>	Were the appropriate processes correctly followed and have the necessary signoffs been gathered?	Did we identify the actual risks to stakeholders and gather the relevant information to effectively manage them?

The biggest obstacles to taking a scientific approach to product development and organizational change are cultural and organizational, as we discuss in (to come). In most cases, organizations have simply never taken an experiment-based approach and lack the skills and experience to implement it. In the context of product development, understanding how to design and execute experiments and analyze data is both hard and critically important, and yet they are not part of the core curriculum in most MBA programs or courses in software design and analysis. In the context of teams operating production systems, we need to run disaster recovery training exercises on a regular basis and build up runbooks for managing these systems so that when disasters strike teams know how to react.

Conclusion

In this chapter we have laid out the foundations of a scientific approach to exploring new work, whether new business models and products, or internal enterprise work such as building new tools or adopting new processes. We have defined exactly what we mean by risk, measurement and uncertainty, and shown how applying the principles and practices of the Lean Startup movement provides a superior way to manage the risks involved in making an investment decisions.

We also presented Boyd’s OODA model, which shows that adaptation to our environment is a continuous and ongoing process — for organizations as much as for people. Our ability to compete is based on creating a common orientation across the organization, and enabling the people doing the work to constantly create and practice new repertoire through a process of experimentation following the scientific method. This enables us to more effectively observe our environment, get inside the OODA loops of other organizations, and act — to better serve our customers and to shape our environment.

Questions for readers:

- How to model investment risk and return in your business plan? How do you create this model and what evidence is it based on?
- What are the variables in the plan with the highest information value? What measurements have been made to reduce the uncertainty in these variables?
- How confident are you that people will find the work you are currently doing valuable? How can you find out?
- Have you tried out the product you are working on with any of its intended users? What did you change as a result?

How to Explore Uncertainty to Capitalize on Opportunity

It was darkness which produced the lamp. It was fog that produced the compass. It was hunger that drove us to exploration.

— VICTOR HUGO

In this chapter we will cover the practices to support the principles discussed in [Chapter 3](#) to explore opportunities in conditions of extreme uncertainty, especially when considering new business models or products. We introduce the concept of *Discovery* to show how to quickly map out a business hypothesis to create a shared understanding of a problem, and engage wider stakeholders from across the organization to get buy-in and alignment to our vision.

We will share concrete tools and techniques to safely create and test hypotheses to solve real business problems that are identified and validated through the execution of our customer development process.

Following that we then describe how to use a disciplined, scientific and evidence-based approach to experimentation to answer the fundamental question—*should* we build it?, not *can* we build it?

We will discuss how to test the riskiest assumptions of our hypothesis and generate empirical data to support our decision to pivot, persevere or stop by creating safe-to-fail experiments using MVPs. Our purpose is to base further investment and portfolio management decisions on evidence, not science-fiction,

building the right thing, at the right time, to execute on opportunities and stop wasting people's time working on ideas that are not valuable.

Discovery

Discovery is a rapid, time boxed, iterative set of activities that integrates the practices and principles from Design Thinking and Lean Startup. We use it intensively at the beginning of the explore phase of a new initiative.

In *Lean UX: Applying Lean Principles to Improve User Experience*, Jeff Gothelf and Josh Seiden state, “Design thinking takes a solution-focused approach to problem solving, working collaboratively to iterate an endless, shifting path toward perfection. It works towards product goals via specific ideation, prototyping, implementation, and learning steps to bring the appropriate solution to light.”¹

When we combine the principle of Design Thinking with Lean Startup practice, we are able to build a continuous feedback loop into our development cycle with real customers. The principle is to invest the minimum amount of effort to provide the maximum amount of learning, and use the outcomes of our experiments to base our decision to pivot, persevere or stop.

During Discovery, we create a collaborative and inclusive environment for a small cross-functional and multidisciplinary team to explore a business, product or improvement opportunity. The team should be fully dedicated and co-located to maximize the effectiveness of real-time decision-making and speed of learning. It must assume ownership of delivery and be empowered to make the necessary decisions to meet the objectives of the initiative.

When forming a team, it is key to keep the group small, including only the competencies required to explore the problem-domain. Large teams are ill-equipped for rapid exploration and learning at the speed required to be successful due to slow decision making and siloed based thinking including conflicting senses of purpose, goals and priorities. The group must know their limitations and boundaries, taking responsibility to reach out and engage others outside the group for input and collaboration when required.

1. (to come), Preface

The final — and too often forgotten — member of the team is the customer. It is easy to fall into the trap of only seeing the customer as a consumer of the solution we have created. In fact, customers are critical stakeholders. Their input is the key ingredient and most objective measure of how valuable our solution is or can be. We view them as a co-creator of value for any solution through the feedback they provide. Their needs must always be the focal point for everything we do.

Creating a Shared Understanding

When you want to build a ship, do not begin by gathering wood, cutting boards, and distributing work, but awaken within the heart of man the desire for the vast and endless sea.

— ATTRIBUTED TO ANTOINE DE SAINT-EXUPÉRY
Citadelle

When starting a new piece of work it is imperative that the group strives to create an environment to maximize the potential of all those involved. People learn, change and improve when involved in a process that is energising, interactive and adaptive based on what they discover by taking part in the process itself.

As Dan Pink argues in *Drive*, there are three key elements to consider when building an engaged and highly motivated team. Firstly, success requires a shared sense of purpose by the entire team. The vision needs to be challenging enough for the group to have something to aspire to, but clear enough that everyone can understand what they need to do. Secondly, people must be empowered by their leaders to work autonomously to achieve the team objectives. Finally, people need the space and opportunity to master their discipline rather than just learning to achieve *good enough*.



Figure 4-1. Building a Shared Understanding of the Vision

The process of shaping the vision begins by clearly articulating the problem statement that the team will try to solve. This essential step is often overlooked

or we assume everyone knows what it is. The quality of a problem statement increases a team's ability to focus on what really matters, and more importantly, what does not. By getting our team to develop a shared understanding of our goals and what we aim to accomplish, we improve our ability to generate better solutions.

TIP

Go Gamestorming

Gamestorming ² by David Gray, James Macanufo and Sunni Brown, and the supporting *Go Gamestorming Wiki* ³ contain numerous games that encourage engagement and creativity while bringing structure and clarity to collaborative ideation, innovation and improvement workshops.

One of the fundamental techniques of Discovery is the use of visual artefacts, models and information radiators to communicate and capture group learnings. Using graphical templates and exercises to externalize ideas helps our team to articulate, debate and evolve concepts and ideas to form consensus. Secondly, it helps to de-personalize and anonymize thoughts, enabling further safety to debate *ideas* over *individuals* — minimizing egos, HiPPOs (Highest Paid Person Opinions) and extroverts attempting to run the show.

Structured Exploration of Uncertainty

If you want to have good ideas you must have many ideas.

— LINUS PAULING

When exploring uncertainty, it is important that we start broad to generate as many as ideas as possible to cycle through before narrowing our focus on where we will start.

lastminute.com is a travel retailer in Europe, operating in a highly competitive industry with major players and new startups aiming to disrupt the travel marketplace everyday. In order to stay relevant, they need to innovate their product faster and smarter than their competitors. They invited customers to become part of the innovation process which included two days running co-creation workshops that generated over 80 new ideas for their online products that were

2. (to come)

3. Go Gamestorming Wiki, <http://www.gogamestorm.com/>

aligned to business goals. The team then set up an innovation lab in a hotel lobby for a week, rapidly experimenting with each idea to discard or validate a viable customer problem to implement. Within days, the team identified three winning ideas to invest further effort in developing — resulting in over a 100 per cent increase in conversion in this product. ⁴

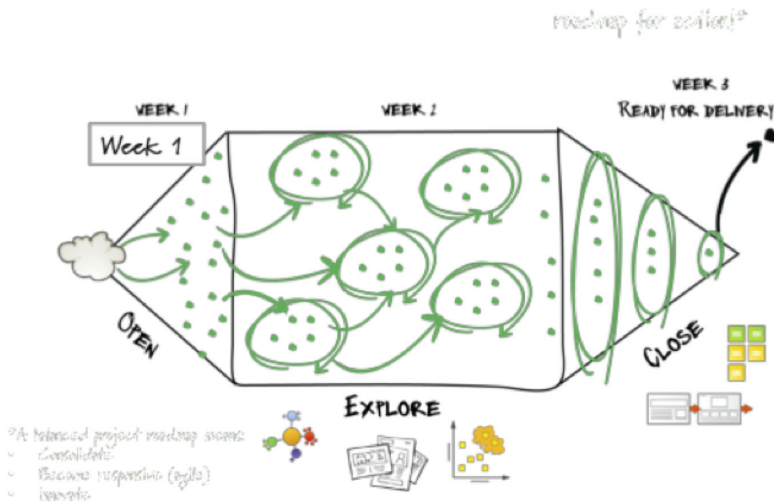


Figure 4-2. Divergent and Convergent Thinking

Divergent thinking is the ability to offer different, unique or variant ideas adherent to one theme, while convergent thinking is the ability to identify the potential solution for the given problem. We start exploration with divergent thinking exercises designed to generate numerous ideas for discussion and debate. We then use convergent thinking to identify a possible solution to the problem. From here, we are ready to formulate an experiment to test it.

What Business Are We In?

Business models are transient and prone to disruption by changes in the external competitive environment, advances in design and technology, and wider social and economic change. Organizations that misjudge their purpose, or cannot sense and then adapt to these changes, will perish.

4. lastminute.com Innovation Lab, <https://www.youtube.com/watch?v=r64rrgbcEHo>

Organizations can be rendered obsolete by competitors that solve the same problem with an alternate or superior offering for the customer. Business definition and identification of future opportunity must be continually challenged and ever evolving. Allowing complacency to sneak in due to current success is the quickest path to failure for tomorrow. We only need to cite examples such as Blockbuster versus Netflix, or HMV and Tower Records versus iTunes, YouTube and Spotify, to illustrate the point that no business model or competitive advantage is indefinitely sustainable.

Winning organizations continually experiment and test theories to learn what works and what does not. As illustrated in (to come), 60%-90% of ideas do not improve the metrics they were intended to improve ⁵, but the ones that do could have a massive impact on the business' future fortunes.

Understanding Our Business Problem, To Inform Our Business Plan

As Steve Blank, author of *The Four Steps To The Epiphany* and *The Startup Owner's Manual* says,

A business plan is the execution document that existing companies write when planning product-line extensions where customer, market and product features are known. The plan is an operating document and describes the execution strategy for addressing these 'knowns.'

The primary objective of a new business initiative is to validate its business model hypotheses (and iterate and pivot until it does). Search versus execution is what differentiates a new venture from an existing business unit. Once a business model is validated, then it should move into execution mode. It's at this point the business needs an operating plan, financial forecasts and other well-understood management tools. ⁶

It is critical to consider many different business models in the early stages of starting a new initiative. We don't want to commit to a plan until we test the

5. Online Experimentation at Microsoft, <http://ai.stanford.edu/~ronnyk/ExPThinkWeek2009Public.pdf>

6. Steve Blank, <http://steveblank.com/2012/03/05/search-versus-execute/>

The *Business Model Canvas* created by Alex Osterwalder and Yves Pigneur, along with 470 co-creators, shown in [Figure 4-3](#), is a simple, visual business model design generator. It is a strategic management and entrepreneurial tool that enables teams to describe, design, challenge, invent, and pivot business models. When compared to writing a business plan, which can become a lengthy process, we can outline multiple possible models on a canvas, each of which should be timeboxed to 30 minutes.



HOW TO EXPLORE UNCERTAINTY TO CAPITALIZE ON OPPORTUNITY

- **Customer segment** — who are we targeting to create value for? Who are our customers?
- **Value proposition** — what problems are we going to solve to create value for our customers?
- **Channels** — through what channels are we aiming to reach out to our target customers?
- **Customer Relationships** — what type of relationship does each of our customers expect us to create and maintain with them?
- **Activities** — what activities will be required to support our value propositions?
- **Resources** — what resources, people, technology and support will be need to all the business to operate?
- **Partnerships** — who do we need to build partnerships with? Who are our key suppliers or could be needed to provide support resources or activities for our value proposition?
- **Cost** — what are the most important inherent cost with our business?
- **Revenue** — for what value are our customer willing to pay? How much and often?

By populating the individual elements of the template, we are prompted to consider any potential idea in terms of the entire business' component building blocks. By populating the entire template we are encouraged to think about how the pieces fit together to support the greater opportunity in a holistic manner. It is key to remember that each component of the canvas represents a set of hypotheses and associated assumptions that require validation to prove that our business model is sound.

Beyond the template itself, Osterwalder also came up with four levels of strategic mastery to competing on business models to reflect to strategic intent of organizations;

- **Level 0 Strategy** — *The Oblivious*: Focus on products/value propositions alone rather than the value proposition AND the business model.

- Level 1 Strategy — *The Beginners*: Use the Business Model Canvas as a checklist.
- Level 2 Strategy — *The Masters*: Outcompete others with a superior business model where every one of the business model building blocks reinforces each other (e.g. Toyota, Walmart, Dell).
- Level 3 Strategy — *The Invincible*: Continuously disrupt themselves while their business models are still successful (e.g. Apple, Amazon.com).

Our ability to recognize what strategy we are pursuing when creating business models is the first step towards creating a shared understanding of what innovation approach will be most effective in helping us achieve our goals.

The primary objective of the Business Model Canvas is to externalize the business hypothesis and make its assumptions clear so we can identify and validate the main risks. The canvas provides a framework to create a shared understanding of each business model, in terms that are understood by all — building a shared sense of ownership and enabling collaboration throughout the organization. The Business Model Canvas differs from other canvases list in Table 4-1 as it doesn’t make the assumption that product-market fit is the riskiest assumption that must be tested first.

There are a number of canvas created by others that focus on product development including;

Table 4-1. *Visual Canvases*

Name	Purpose
<i>The Lean Canvas</i> ^a	Makes the assumption that product-market fit is the riskiest assumption that must be tested.
<i>The Opportunity Canvas</i> ^b	To drive first discussions about what we’re building and why. Then understand how satisfying those specific customers and users furthers the organization’s overall strategy.
<i>Value Proposition Canvas</i> ^c	Describe how our products and services create customer gains. How do they create benefits our customer expects, desires or would be interesting in using.

^a The Lean Canvas, <http://www.leancanvas.com>

^b The Opportunity Canvas, <http://comakewith.us/tag/opportunity-canvas/>

^c Value Proposition Canvas, http://www.businessmodelgeneration.com/downloads/value_proposition_canvas.pdf

Understanding Our Customers

The single most important thing to remember about any enterprise is that there are no results inside its walls. The result of a business is a satisfied customer.

— PETER DRUCKER

In order for any product or solution to be successful customers must want to use it, and indeed pay money for it. For a team to build a solution that addresses a customer problem or need, it is essential the team must understand whom we are trying to reach and why we are targeting them.

Put A Face To Your Customer

A *persona* is a representation of the problems, needs, goals and behavior of a hypothesized group of customers. Personas are based on relevant information and insight known to the creators, they are essentially a collection of assumptions that must be tested and refined throughout our customer development process.

When creating a persona it is important to remember the following points; * Define and brainstorm initial persona very quickly to get alignment across the team * Iteratively redefine persona based on evidence from user research, testing and feedback during the customer development cycle * Continually realign persona and the business/product vision as the product starts to emerge

Personas are just a starting point we can use to create a shared understanding of our customers. They are never truly objective or empirical; that is not their purpose. We use personas to create empathy with our customers, and move the conversation from what our own individual preferences may be, to focus on what the selected persona would perceive to be valuable.

Having empathy for customers' situation is a powerful force. When we empathize, we enhance our ability to receive and process information ⁷. Empathy in design requires deliberate practice. We must design experiments and interaction opportunities to connect with our customers in meaningful ways and chal-

7. IDEO: Empathy on the Edge, http://www.ideo.com/images/uploads/news/pdfs/Empathy_on_the_Edge.pdf

lenge our own assumptions, preconceptions and prejudices. We need to assume the role of an interested inquirer, trying to understand the challenges our customers experience.

Creating a balance between empathizing with an experience and analyzing the situation allows us to understand our customers' feelings and perspectives. We can then use that understanding to guide our identification of solution hypotheses and commence the experimentation process.

TIP

Go, Look, See

The design company IDEO ⁸, famous for creating the original Apple mouse, runs workshops in which teams completely immerse themselves in the context in which the envisioned product or service will be used. Their developers read everything of interest about the markets, observe and interview future users, research offerings that will compete with the new product, and synthesize everything that they have learned into pictures, models, and diagrams. The result is deep insights into customers that are tested, improved, or abandoned throughout the iterative development process.

At Toyota, *genchi genbutsu* is designed to allow leaders to identify existing safety hazards, observe machinery and equipment conditions, ask about the practiced standards to gain knowledge about the work status, and build relationships with employees. The objective of *genchi genbutsu* is take *gemba* walks to understand the value stream and its problems rather than review reports or make superficial comments.

Similarly, *getting out of the building* (a phrase popularized by entrepreneur and author Steve Blank), is a customer development technique used to get feedback and focus early product development efforts around early adopters through qualitative inquiry.

People who cannot temporarily let go of their role and status, or set aside their own expertise and opinions, will fail to be empathic with others who have conflicting thoughts, experience, or mental models. Therefore, the ability to listen and ask the right questions becomes a powerful skill to develop a deep insight that is the foundation of effective problem solving and experimentation.

8. IDEO, <http://www.ideo.com>

Turning Insights and Data into Unfair Advantage

The ability to discover and leverage critical insight is essential to high performing organizations. Previously, we operated in a relatively small data universe, as there was a high cost associated with the collection, storage and processing of data. The big data movement has provided technologies and techniques for reviewing, processing and correlating large existing data sets. Organizations can seek to gain additional value from insights into how and why customers are interacting with the product and solutions. We can detect weak signals that tell us what is working well — or not so well — and use that information to improve existing services or create new offerings. When combined, software, analytics and data form a key pillar in our organization’s intellectual capital.

Access to, and understanding of, existing customers is a significant competitive advantage established organizations have over startups. Startups have the challenge of trying to gain market reach and traction due to the lack of access and availability of known customer data. On the other hand, established organizations have an existing market and customer data that can be reused and leveraged to unearth new opportunities.

Organizations are now able to form hypotheses such as, “Why are customers cancelling their memberships?” or “How are customers related to one another?”, and quickly create inexpensive experiments to test their hypotheses based on existing data. This is a powerful technique to remove decision bias from our prioritization process, and enable data-driven decisions.

Data analytics enables us to invert the discovery process by retrospectively looking at how customers are using existing services, and do forward projections for opportunities to create new business models, products or services.

NOTE

How Companies Mine Data To Discover Your Secrets

In *The Power of Habit*, Charles Duhigg writes, “almost every major retailer, from grocery chains to investment banks to the U.S. Postal Service, has a ‘predictive analytics’ department devoted to understanding not just consumers’ shopping habits but also their personal habits, so as to more efficiently market to them.”

Target notably used this data to particularly creepy effect in order to identify and market to pregnant women. One of the consequences of being pregnant is a need to prepare for your new child through buying lots of stuff. Target wanted to encourage pregnant families do most of their shopping at Target, potentially capturing them as major customers for life. They analyzed their existing data on customers to find a way to identify women in their second trimester of pregnancy, so they could be targeted for offers.

Target was able to identify changes in buying patterns for 25 key products, including nutritional supplements, cotton balls and unscented lotion, that accurately predicted not only pregnancy but also due date. As a result they were able to send pregnant women relevant coupons — cleverly disguised amongst other vanilla offers so the women didn’t realize they were being targeted — to encourage them to do their pre-baby shopping at Target ⁹.

Big data is a tool, not a solution, and crucially it does not replace empathy. We still require human intuition and innovation to improve the problem definition and identify customer needs and problems, which allows us to form hypotheses that can be tested against the data. Cross-functional teams, personas and user interviews are all powerful tools that enable us to design experiments more effectively and rapidly. We need to learn how to listen and learn from data through unbiased analysis, otherwise the data collected is useless: “Data, like a flashlight, is only as useful as the person wielding it and the person interpreting what it shows.” ¹⁰

Using Insight To Inform Hypotheses and Experiments

During Discovery numerous members of the cross-functional team will have (and should be encouraged to share) interesting and valuable insights into the organization, customers, business, channels or markets. By sharing these in-

9. <http://www.forbes.com/sites/kashmirhill/2012/02/16/how-target-figured-out-a-teen-girl-was-pregnant-before-her-father-did/>

10. Scott Berkun, <http://scottberkun.com/2013/danger-of-faith-in-data/>

sights with the team we can generate new perspectives and inspiration for new products or solutions gleaned from the data.

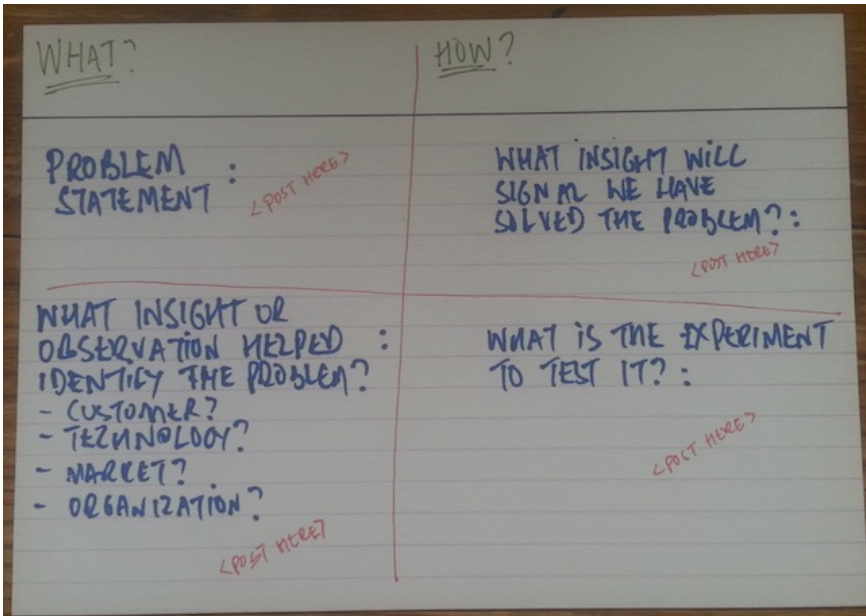


Figure 4-4. A problem statement canvas

Ask those involved in Discovery to share what interesting insights and data they have observed to inform, create or challenge problem statements based on a number of perspectives, for example;

- **Customer** - What specific information does the group have about existing customers? What are their usage and engagement behaviors? How can those insights help to shape future opportunities within existing product offers?
- **Market trends** - Industry trends of the market we are attempting to enter is key to understanding how and where opportunities exist i.e. mobile technology, location based services, mobile payments. What are the market trends for the product we are creating? How do we measure against them?
- **Organization** - What specific information does the group have about our organization? Where is the organization focusing its efforts? What are the

impacts of those efforts? How much of the wider competitive landscape does it cover? Where is the organization most effective?

- **You Will Not Believe!** - Every company has individuals that are willing to share interesting and astounding facts about the business or customer base. How can we start to test if they are indeed true and/or offer opportunities to create a new value proposition as a result?

By making this information visible and discussing it, we can attempt to identify new business models and value propositions appropriate for the business, given its current constraints and identified problem statements.

Accelerate Experimentation with MVPs

The Lean Startup movement challenges the assumption that customers must have all imaginable features available in a product before they will start to use it. Eric Ries coined the term *Minimal Viable Product* (MVP) to describe a strategy directed toward investing the minimal amount of resources to test the underlying assumptions of our hypotheses with customers. The objective is to eliminate the waste generated by over-engineered solutions, and accelerate our learning through testing a solution with early customers as soon as possible.

A MVP enables us to use the minimum amount of effort, to generate the maximum amount of learning when experimenting with customers. The goal of using a MVP is to execute an experiment that tests the assumptions of our hypothesis as cheaply, quickly and effectively as possible, in order to learn if our solution addresses the customer problem we have identified. It eliminates the parts of the solution hypothesis that create unnecessary complexity and use of resources when experimenting with our initial targeted customers. The outcome of the experiment is learning, which enables us to make an evidence-based decision to pivot, persevere or stop our current hypothesis.

An important distinction to understand is the difference between the MVP and the initial release or beta test of actual products and solutions to customers — as is probably the most commonly misunderstood term in Lean Startup movement. Marty Cagan, author of *Inspired: How to Create Products Customers Love* and ex-SVP for eBay, differentiates between a product release and a MVP test, a logical separation that helps when we are trying to decide exactly what we are going to release and sell to early customers. While Ries' definition focuses on testing and learning, Cagan's focuses on discovering the core features of what we

wish to create and how we bring them to market. To this end, he focuses on three critical characteristics or measures; “people choose to use it or buy it; people can figure out how to use it; and we can deliver it when we need it with the resources available – also known as valuable, usable and feasible.”

Should We Build It, Not Can We Build It?

JustGiving is an online fundraising platform that has raised over £2 billion GBP for charities. JustGiving wanted to explore new business models to fund community initiatives that are not necessarily affiliated with a charity.

They formed a small co-located team to rapidly experiment with customers, running sessions with a prototyped version of a crowd-funding platform, complete with real community projects that were seeking support. Based on the positive reaction from customers, they proceeded to create a concierge MVP, launching the same trusted community projects with real customers while manually handling back office tasks such as project setup, payment processing and collection, to see how the product would perform in the market.

Within 7 weeks from the start of the initiative, JustGiving had validated a repeatable business model that they could start to scale to a business in its own right. The product has now become YIMBY ¹¹ with success stories that include purchasing basketball wheelchairs for teams, tools to expand a community gardens, and saving 140 year old Kettering Town Football Club.

MVPs do not guarantee success; they are designed to test the assumptions of a problem we wish to solve without over investing. By far the most likely outcome is that we learn our assumptions are invalid and we need to pivot or stop our approach. Our ultimate goal is to minimize investment when exploring solutions until we are confident we have discovered the *right product*, then exploit the opportunity by adding further complexity and value to build the *product right*.

11. Yes In My Back Yard, www.yimby.com

Table 4-2. *An example set of types of MVPs*

Type	Pros	Cons	Examples
Paper	Speed, Visual, Creates shared understanding	Limited interaction, Does not test usability or hypothesis	Diagrams, Wireframes, Sketches
Interactive Prototype	Test design and usability, iterate solutions at speed, Test qualitative customers interviews	Does not test hypothesis or supporting technology	HTML or clickable mock-ups, Videos
Concierge	Reduced complexity, Supports generative research, Validate assumptions qualitative with a small investment	Limited scalability, manual, resource intensive, Customer is aware of human involvement	AirBnB founders offering air beds to customers during Democratic National convention, Collision installation for Stripe ^a
Wizard of Oz	Working solution from customer perspective, person playing the role of the wizard can gain valuable insights from the close involvement, Enables evaluative research for price points and validation of value proposition	Limited scalability based on higher commitment of resources, person playing the role of the wizard must appreciate the functionality of the proposed solution, Difficult to evaluate systems with a large graphical interface component	Tony Hsieh purchasing shoes for initial customers of Zappos.com
Micro-niche	Highly focused test, dedicated to any specific topic, minimal effort	Financial investment to drive traffic, Competition for keywords and customer click throughs	http://whatkatewore.com/
Working software	Test hypothesis in real environment, validate assumptions qualitative	Expensive, scale of relative investment in people and tools	A/B testing, conversion funnels, referral optimization

^a Paul Graham, <http://paulgraham.com/ds.html>

How do our Vision and MVP Work Together?

Cagan stresses the importance of how vision and MVP are intimately related but not the same. Cagan defines vision as the shared understanding that “describes the types of services you intend to provide, and the types of customers you in-

tend to serve, typically over a 2-5 year timeframe.”¹² As such, it provides the roadmap and context for MVPs, and we should be prepared to create many MVPs as we search for a repeatable, scalable customer development process that is aligned with our vision.

Early evangelists, particularly in enterprises, should buy into our entire vision, not just our first MVP experiment. They will need to hear what our organization plans to deliver over the next 6 to 18 months. They are able to fill in the gaps in our solution as they feel the pain of the problem we are trying to solve, and are brought into the vision of what we are trying to achieve. By creating a taste of the solution we are aiming to build, we are giving them an opportunity to react, see evidence that it works and provide feedback on the solution we are building.

Leveraging business interaction and engagement is very important at the early stages of a new initiative. Feedback and evidence gained through the use of a MVP provides better customer insight and learning than aggregate measures of success such as total revenue or total transaction value. The MVP allows us to focus on the right thing to build and provides valuable information on how to evolve, adapt and pivot to meet customer needs discovered through experimenting with them.

12. Marty Cagan, <http://svpg.com/product-market-fit-vs-product-vision>

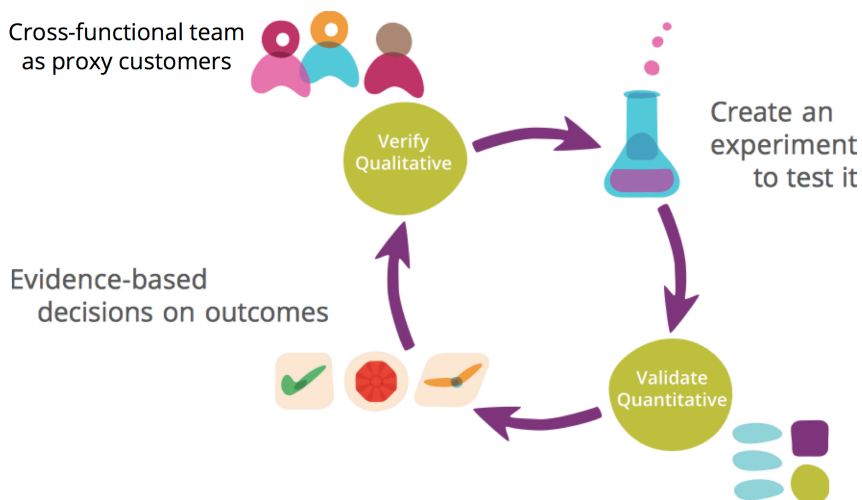


Figure 4-5. *Verify Qualitatively, Validate Quantitatively*

By starting with the question of what we want to learn from the experiment, we can define how we will observe and measure it, and finally devise ways to create the cheapest, quickest and simplest MVP to test our assumptions, measure the effect and use that learning to formulate next steps.

A fundamental point with new initiatives is the need to preserve cash and iterate rapidly while teams are testing hypotheses to identify a *repeatable* solution. Once those fundamentals are understood, and we achieve a ‘product/market fit’, cash preservation becomes less important than spending and we can start to create a *scaleable* solution.

The One Metric That Matters

When designing MVPs to experiment, it is important to identify one key metric that will tell us if the assumptions in our hypothesis are valid or invalid. For this reason, *Lean Analytics* authors Alistair Croll and Benjamin Yoskovitz introduced the concept of *One Metric That Matters* (OMTM). OMTM is a single metric that we prioritize as the most important to drive decisions depending on the stage of our product lifecycle and business model. It is not a single measure that we will use throughout our product lifetime: it will change overtime depending on the problem area we wish to address.

We focus on One Metric That Matters to:

- provide information to answer the most pressing question we have by linking it to the assumptions in the hypothesis we want to test;
- create focus, conversation and thought to identify problems and stimulate improvement;
- provide transparency and a shared understanding across the team and wider organization;
- support a culture of experimentation by basing it on rates or ratios not averages or totals relevant to our historical dataset

It should not be a lagging metric such as Return on Investment (ROI) or customer churn, both of which measure outputs after the fact — lagging indicators become more interesting later when we have achieved product/market fit. By initially focusing on leading metrics we can get an indication of what is likely to happen, thus address a situation quicker to try and change outcomes going forward. For example, customer complaints is often a leading indicator of churn. If customer complaints are rising, we can expect that customers will leave and churn will increase. Our OMTM should always evolve as we learn more about the problem we want to solve.

The purpose of the OMTM is to objectively gain evidence that the changes we are making to our product is having a measurable impact on the behaviour of our customers. Ultimately we are seeking to understand;

- Are we making progress (what)?
- What caused the change (why)?
- How do we improve (how)?

Founder of Intuit Scott Cook says that founders should focus on “love metrics”, meaning how much people love the product, how often they come back and how delighted they are in the early stages. “If you’re not getting high activity from the users you already have, it’s time to pivot.”

TIP

Use A3 Thinking as a systematic method for realizing opportunities for improvement

A3 Thinking is a logical problem-solving tool to capture critical known information and define the focus and constraints of the team. Later, it becomes the measure to test our outcomes against. An A3 report (so called because it fits on a piece of A3 size paper) is composed of seven elements, and embodies the Plan-Do-Check-Act cycle of experimentation;

- **Background** capture critical information to give understanding of the extent and importance of the problem. Tying the background to the goal statement reduces opportunity of waste by focusing on wrong areas
- **Current condition and problem statement** the problem the business stakeholder wants to address in simply understandable terms. Not a 'lack of solution' statement i.e. our problem is we need a Content Management System.
- **Goal statement** how will we know that our efforts were successful at the end of our implementation? Ideally we will have one key metric for success. For example: "Our goal is to reduce system failures when compared with the previous test results of 22 major issues". Our target is to reduce this by 20%.
- **Root-cause analysis** detail hypothesis and assumptions or set of experiments performed to test for cause and effect.
- **Countermeasures** the steps of an experiment to be implemented to test hypothesis
- **Check/confirmation effect** define a method for assessing if the countermeasures have had an effect or not
- **Follow up actions and report** identify further steps and share learning with team and wider organization

For more on A3 thinking read *Understanding A3 Thinking: A Critical Component of Toyota's PDCA Management System* by Durward K. Sobek II. and Art Smalley. Other examples including Elevator Pitch ¹³ and 5W's and 1H [Who, What, Where, When, Why and How]

Remember, metrics are meant to hurt, not make us feel good that we are winning. They must be actionable and trigger a change in our behaviour or understanding. We need to consider these two key questions when deciding on what our OMTM will be:

What is the problem we are trying to solve?

13. <http://www.gogamestorm.com/?p=125>

- *Product development* - are we attempting to create new product or services that involve customers? How will we know we are engaging them and they are interested in our product?
- *Tool selection* - are we attempting to select a tool for usage in the organization? How will we know it is the best tool for the process we want to use?
- *Process improvement* - are we attempting to improve our internal capability and efficiency? How will we know if our changes are having the desired impact?

What stage of the process are we at?

- *Problem validation* - are we trying to identify a problem exists by talking to people to see if they are experiencing the pain of the issue we're trying to solve?
- *Solution validation* - are our people demonstrating alignment and buy-in for the problem we are aiming to solve through qualitative interviews?
- *MVP validation* - are we creating experiments to quantitatively prove that our solution is working to solve the problem statement we have identified?

OMTM is a helpful tool for simplifying the complexity of analytics and specifically tells us if our solution is succeeding or not. Once we have defined the key metric to focus our efforts, we can identify supporting metrics that provide insight into other areas and support decision making.

A good example of OMTM is at LinkedIn, the team does not talk about “total page views”, but instead “profile views” — the number people using LinkedIn to search for and find other people, and the number of profiles on LinkedIn being viewed.¹⁴ Similarly for users of LinkedIn, “Who’s viewed my profile?” provides a quantitative measure that your profile has been viewed by X number of people in the last Y days. If X is decreasing as Y is constant you’re losing viewers.

14. <https://medium.com/what-i-learned-building/ab24a585b5ea>

Conclusion

In this chapter we have shared practices to enact the principles that allow us to safely explore opportunities in conditions of extreme uncertainty — especially new product development and business model innovation. We've introduced *Discovery*, including concepts and tools that allow us to invest the minimum amount of effort to obtain the maximum amount of learning to make measurable progress towards exploiting opportunities.

A key outcome of *Discovery* is creating and developing a shared understanding within teams, and communicating this to rest of organization. This is achieved by being transparent and making work visible. Showcase and go beyond the product and technical team to engage the entire organization using the artefacts we create: Problem Statements, Business Model Canvas, Persona and A3 reporting — seek wider collaboration to encourage feedback, alignment and buy-in.

Finally, we must adopt the mindset that all our ideas are hypotheses that contain underlying assumptions that must be tested, and most of these will prove to be wrong. By basing our decision-making on information gleaned from fast, inexpensive experiments using MVPs, we make better decisions on further investment and portfolio management. Ultimately informing options to pivot, persevere or just stop engaging in wasteful activities.

Questions for readers:

- *Discovery* is designed to identify and validate a real customer problem and solution hypothesis on how to solve it, using experiments that test the underlying assumptions of the hypothesis. What is your current business hypothesis and how would you create an experiment using a MVP to test it?
- Continually address the question "*should* we build it, not *can* we build it?". This is answered by basing decisions on information obtained from safe-to-fail experiments involving our customers. What learning outcomes would your team use to decide when to pivot, persevere or stop?
- When innovating, select *One Metric That Matters* to simplifying the complex nature of analytics and to tell us if our MVP is succeeding or failing. It should be based outcomes that are aligned to our strategy and not vanity metrics related to volume of work we have done. What is yours?

Finding Product-Market Fit Before Starting To Scale

The Edge... there is no honest way to explain it because the only people who really know where it is are the ones who have gone over it.

— HUNTER S THOMPSON

In this chapter we will discuss how to identify when product-market fit has been achieved, to exit explore and start to exploit our product with its identified market. We'll share how to use customized metrics to understand if we are achieving measurable business outcomes while continuing to solve our customers' problems by engaging them throughout our development process.

We cover how organizations set themselves up for success with the right strategy, structure and support to find both internal and external customers to provide valuable feedback and insight as we grow our product. We will address how to leverage existing capabilities, services and practices to scale our product while seeking internal advocates within our organization to collaborate with. Finally we'll describe the metrics and engines for growth to manage transition between business model horizons as we start to scale our solution.

Innovation Accounting

It is not enough to do your best; you must know what to do, and then do your best.

— W EDWARDS DEMING

We live in a world of data overload, where any argument can find supporting data if we are not careful to validate our assumptions. Finding information to support a theory is never a problem, but testing the theory and then taking the correct *action* is still hard.

As discussed in [Chapter 3](#) the second largest risk to any new product is building the wrong thing. Therefore, it is imperative we don't over invest in unproven opportunities, doing the wrong thing the right way. We must begin with confidence that we are actually doing the right thing. So how do we test if our intuition is correct, especially when operating in conditions of extreme uncertainty?

Eric Ries introduced the term '*Innovation Accounting*' to refer to the rigorous process of defining, experimenting, measuring and communicating the true progress of innovation for new products, business models or initiatives. To understand whether our product is valuable and hold ourselves to account, we focus on obtaining admissible evidence and plotting a reasonable trajectory while *exploring* new domains.

Traditional financial accounting measures such as operating performance, cash flow or profitability indicator ratios like return on investment (ROI) are not designed for innovation and often have the effect of stifling or killing new products or initiatives. They are optimized, and more effective, for *exploiting* well understood domains or established business models and products. By definition, new innovations have a limited operating history, minimal to no revenue, and require investment to start up. In this context, return on investment, financial ratio analysis, cash flow analysis and other well established practices provide little insight on the value of new innovation or enable investment evaluation against the performance of other well established products through financial data comparison alone.

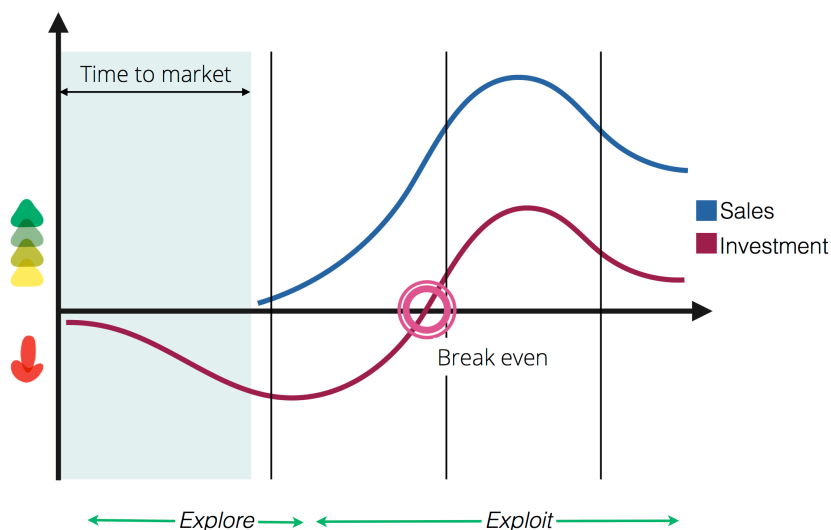


Figure 5-1. *Investment to Sales Ratios for Early Stage Innovations*

When *exploring*, accounting must not be ignored or deemed irrelevant. It simply needs to be interpreted differently to measure the outcomes of innovation and early-stage initiatives. Our principles of accounting and measurement for innovation must address the following aspects:

- Establish accountability for decisions and evaluation criteria
- Manage the risks associated with uncertainty
- Signal emerging opportunities and errors
- Provide accurate data for investment analysis and risk management
- Accept that we will at times, need to move forward with imperfect information
- Identify ways to continuously improve our organization's innovation capability

WARNING

Measurement Fallacy

‘What you measure is what you get’ - Kaplan and Norton ¹

One of the key ideas of Eric Ries’ *Lean Startup* was the use of actionable metrics. He advocates that we should invest energy in collecting the metrics that help us make decisions. Unfortunately, often what we tend to see collected and socialized in organizations are vanity metrics — designed to make us feel good, but offer no clear guidance for what *action* to take.

In *Lean Analytics*, Alistair Croll and Benjamin Yoskovitz note, “If you have a piece of data on which you cannot act, it’s a vanity metric... A good metric changes the way you behave. This is by far the most important criterion for a metric: what will you do differently based on changes in the metric?” ²

Table 5-1. *Examples of Vanity vs. Actionable Metrics*

Vanity	Actionable
Number of visits: Is this one person who visits a hundred times, or are a hundred people visiting once?	Funnel metrics, cohort analysis: define the steps of our conversion funnel, then group users and track their usage lifecycle over time.
Time on site/number of pages: These are a poor substitute for actual engagement or activity unless your business is tied to this behaviour. They address volumes, but give no indication if the customer can find the information they need.	Number of sessions/user: Define overall evaluation criterion for how long it should take for a session (or action) to complete on the site, then measure how often the user is performing it successfully
Emails collected: A big email list of people interested in a new product is exciting but until we know how many will open our emails (and act on whats inside them), this isn’t useful.	Email action: Send test emails to a number of registered subscribers and see if they do what we tell them to do.
Number of downloads: While it sometimes affects your ranking in app stores, downloads alone don’t lead to real value.	User Activations: identifies how many people have downloaded the application and used it. Account creations and referrals provide more evidence of customer engagement.

1. The Balanced Scorecard—Measures that Drive Performance, p70, <http://hbr.org/1992/01/the-balanced-scorecard-measures-that-drive-performance>
2. (to come), p13

Vanity	Actionable
Tool usage: Level of standardization and reuse in the enterprise tool chain	Tooling effect: Cycle time from check-in to release in production for new line of code.
Number of trained people: Number of people that have been through Agile training and successful obtained certification	Net Promotor Score: Responses of trainees that attended Agile training and want to implement it in daily work

3 4 5 6

In *How to Measure Anything*, Douglas Hubbard recommends a good technique for deciding on a given measure: “If you can define the outcome you really want, give examples of it, and identify how those consequences are observable, then you can design measurements that will measure the outcomes that matter. The problem is that, if anything, managers were simply measuring what seemed simplest to measure (i.e., just what they currently knew how to measure), not what mattered most.” ⁷

By combining the principle of actionable metrics with Hubbard’s recommendation for how to create the measures that matter most, we can go beyond traditional internal efficiency and financial measurement to focus on value from the perspective of the stakeholders that matter most, our customers.

3. Lean Analytics, <http://shop.oreilly.com/product/0636920026334.do>
4. Ash Maurya, Funnel metrics, cohort analysis, <http://practicetrumpstheory.com/2010/07/3-rules-to-actionable-metrics/>
5. Dan McClure, Pirate Metrics, <http://500hats.typepad.com/500blogs/2007/09/startup-metrics.html>
6. A/B Testing and Controlled Experiments, <http://www.infoq.com/presentations/controlled-experiments>
7. (to come), p37

Stakeholder	Measure	Current	Target	Trend
Customer	% users that complete sales flow	30%	45%	▲
	% retention	20%	25%	▲
	Net Promoter Score	44	60	▲
Business	% visits to sign up for service	20%	25%	◀▶
	% conversion to paying customers	15%	20%	▲
	Customer acquisition costs	\$0.25	\$0.05	◀▶
	Life time customer vale	\$0.30	\$0.80	▼
	% attrition	30%	15%	▼

Figure 5-2. *Lean Enterprise Example Innovation Scorecard*

Customer success metrics are measures provide insight on whether customers believe our product to be valuable. Business metrics on the other hand, focus on the success of our own business model. As we noted before, collecting data is never an issue for new initiatives, the danger lies in getting lost in all the noise.

For this reason, we continue to use the practice of *One Metric That Matters* to prioritize and drive decisions based on the stage of our product lifecycle and business model. As ever, it is not a single measure that we will use throughout our product lifetime, it will change overtime depending on the problem area we wish to address.

To help us improve, our dashboard should only show metrics that will trigger a change in behaviour, are customer-focused, and present targets for improvement. If we are not inspired to take action based on the information presented on our dashboard we are measuring the wrong thing, or have not drilled down enough to the appropriate level of actionable data.

WARNING

Statistical Paralysis and Expert Opinion

During the early stages, it is critical that we worry less about statistical significance and focus on creating experiments designed to test our subjective intuition and assumptions. As Hubbard notes, “the concept of statistical significance is vastly overused by those who don’t quite understand what it means”.

Initially, it is acceptable to use gross measures. Does this make things better, worse or the same? If it looks close, then we can experiment to determine the statistical difference and decide if we want to continue pursuing the idea.

When *exploiting* initiatives, traditional measurement and control become more relevant. By experimenting we start to gain more knowledge and historical data, which results in improved predictability and forecasting of the direction we want to go. Typically our point of inflection comes when we have found a repeatable business model that we are confident will scale.

Energizing Internal Advocates in the Enterprise

Innovation in large, bureaucratic organizations is challenging because they are inherently designed to support stability, compliance and precedence over risk taking. Leaders that have risen to the top have so because they have worked the system as it has existed to date. Therefore, we need to be careful that any critiques do not become focused on individuals or their behaviour within the system. We need to seek out collaborators and co-creators across the organization without causing alienation, to gain further support for our efforts and cross the chasm to the next stage of adoption curve within the organization. Ultimately, we will need to identify activators in the areas that we need to change to be successful. The best way to encourage the engagement of these groups is *demonstrable evidence* that our efforts are achieving measureable business outcomes.

Without doubt there are people in our organization that are frustrated and curious for change. However they seek safety, context and cover to act before they are willing to become champions of an initiative. Energizing and engaging these people is key. As they become early adopters of our ideas and initiatives, they provide a feedback loop that enables us to iterate and improve our product. They are also our sponsors within the wider organization. In bureaucratic environments, people tend to protect their personal brand and not back the losing horse. Our goal is to give them the confidence, resources and evidence that encourages them to be advocates for our initiative throughout the organization.

Do Things That Don't Scale

Even when we have validated the assumptions of our hypotheses, or achieved what we believe to be product/market fit, it is important that we continue to focus on the same principles of simplicity and experimentation. We must continue to optimize for learning and not fall into simply delivering features. The temptation once we achieve traction is to seek to automate, implement and scale everything identified as “requirements” to grow our solution. However, this should not be our focus.

In the early stages we must spend less time worrying about growth, and focus more on significant customer interaction. We may go so far as to only acquire customers individually — too many customers too early can lead to a lack of focus and slow us down. We need to focus on finding passionate early adopters to continue to experiment and learn with. Then seek to engage similar customer segments to eventually *cross the chasm* to wider customer acquisition and adoption.

This is counterintuitive to the majority of initiatives in organizations, as we are programmed to aim for explosive growth, and doing things that don't scale doesn't fit with what we have been trained to do. Secondly, we tend to measure our required level of service, spend and success in relation to the revenue, size and scope of more mature products in our environment or competitive domain.

We must remember that we are still in the *formulative* stage of our discovery process, and don't want to over invest and commit to a solution too early. We continually test and validate the assumptions from our business model through market experiments at every step. If we have identified one key customer with a problem and can act on that need, we have a viable opportunity to build something many people want. We don't need to engage every department, customer segment or market to start. We just need a focused customer to co-create with. We don't need to build something extraordinary initially, we just have to put in extraordinary *effort* to ensure we are meeting our customers' needs.

Once leaders see evidence of rampant growth when we're operating with unscalable processes, we'll easily be able to secure people, funding and support to build robust solutions to handle the flow of demand. Our goal should be to create a *pull* system for customers that want our product, service or tools, not *push* a

mandated, planned and baked solution upon people that we must *sell* or require them to use (perhaps as part of a “standardization” initiative).

Customer Intimacy

By deliberately narrowing our market to prioritize quality of engagement and feedback from customers, we can build intimacy, relationships and loyalty with our early adopters. People like to feel part of something unique, special and involved from the start.

The Royal Pharmaceutical Society knew that their clinical drug database was the best in the world. They also knew that there must be many more uses for this than just a stack of printed books. But where should they start? Instead of guessing, or building an expensive platform for products, or trying to sign a deal without a product, they used their other major asset: a building full of pharmacists. Through rapid prototyping, user testing with pharmacists working for the society and product research with nearby pharmacies they were quickly able to focus on an app to check for potential interactions between prescribed drugs. There are huge opportunities in licensing the data for international use. By starting with an app that they themselves would use, they were able to understand what international customers might want and also build a great marketing tool.

By keeping our initial customer base small, not chasing vanity numbers to get too big too fast, we force ourselves to keep it simple and maintain close contact with our customer every step of the way. This allows teams more time with customers to listen, build trust and have capacity to insure early adopters that we’re ready to help. Remember, big numbers doesn’t equal big win — meeting unmet needs and delighting customers does.

Build A Runway Of Questions, Not Requirements

The instinct of product teams once product/market fit is achieved is to start building all the requirements of a scalable, fully functioning and complete solution based on the gaps in their MVP. The danger with this approach is that it prevents us from evolving the product based on feedback from customers.

In the early stage we are still learning, not earning. Therefore it is important that we do not limit our options by committing time, people and investment to build

features that we are unsure will produce the desired customer outcomes. We must accept that everything is an assumption to be tested, and continually seek to identify our area of most uncertainty, and formulate experiments to learn more. To hedge our bets with this approach, leverage things that don't scale — building a runway with scenarios planned out as to how we will continue to build out our product.

Our runway should be a list of hypotheses to test, *not* a list of requirements to build. When we reward our teams for their ability to deliver requirements only, it's easy to rapidly bloat our products with unnecessary features — leading to increased complexity, higher maintenance costs and limited ability to change quickly. Features delivered are not a measure of success, *business outcomes* are. Our runway as a series of questions that we need to test to reduce uncertainty and improve our understanding of growth opportunities.

Create a Story Map to tell the narrative of the runway of our vision

Story Maps are tool developed by Jeff Patton explained in his book, *User Story Mapping*⁸ As Patton states, “our software has a backbone and a skeleton - and your map shows it”.

8. User Story Mapping, <http://shop.oreilly.com/product/0636920033851.do>

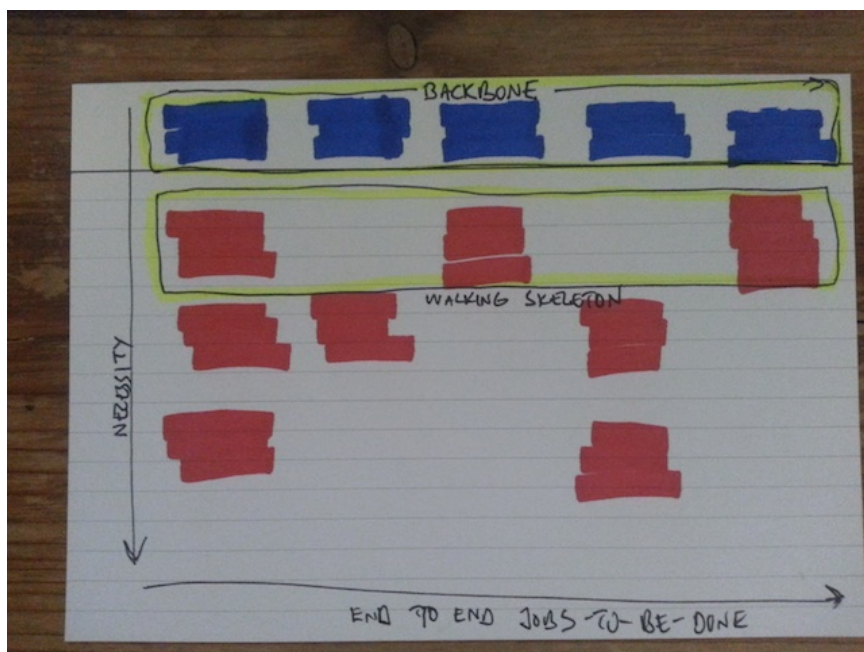


Figure 5-3. *Story maps*

Story maps help with planning and prioritizing by visualizing the solution as a whole. Story mapping is not designed to generate stories or create a release plan — it is about understanding the customer objectives and jobs-to-be-done. Story maps provide an effective means to communicate the narrative of our solution to engage the team and wider stakeholders and get feedback. By going through story maps and telling the story of the solution, we ensure that we have not missed any major components. At the same time, we maximize learning by identifying the next riskiest hypothesis to test while minimising waste and over-engineered solutions that do not fit customer needs as defined in our MVP.

When we start to harden, integrate and automate our product it impacts our ability to rapidly adapt to what we are discovering, often limiting our responsiveness, pace to change and optionality. At this point, we must continuously reset our assumptions and try to restrict integration and automation by leveraging existing services, capabilities or manually processes to deliver our product. Our aim is not to remove ourselves from the customer. We want to ensure that we are constantly interacting together by design. If we optimize for simply

building without constantly testing our assumptions with our customer along the way, we can miss key pain points, experiences and successes, and that is often where the real insight are.

If we want to learn, we have build empathy for our customer and experience their pain. Automating problems may teach us how to be build a scalable product, but it won't teach us about our customers. When we find a customer with a problem and we can solve it manually, we do so for as long as possible. When our customers' quality of service is compromised or we cannot handle the level of demand, we consider introducing automation to address bottlenecks that have emerged through increased use of the product.

NOTE

Leverage Frugal Innovation

Unscalable techniques and practices are not only a necessity — they can be a catalyst for change in a organization culture. Proving it is possible to test our ideas quickly, cheaply and safely offers others in the organization encouragement and confidence that experimentation is possible, resulting in a lasting change to our culture for the better.

Engineering Practices For Exploring

In general, we favor the principles of Toyota Production System by “building quality in” to software, discussed at length in (to come). However, when exploring, there is a tension between the need to experiment and build MVPs, and building in high levels of quality through practices such as test automation.

When we start work validating a new product idea, or validating a new feature in an existing product, we want to try out as many ideas as fast as possible. Ideally we will do this without writing any software at all. But for the software we do write, we don't want to spend a ton of time building acceptance tests and refactoring our code. We will (as Martin Fowler puts it) deliberately and prudently accumulate technical debt in order to run experiments and get validation.⁹

However if our product is successful, we will hit a brick wall with this approach. Changes will become onerous and timeconsuming and the produt will become infested with defects and suffer from poor performance. (perhaps a year

9. <http://martinfowler.com/bliki/TechnicalDebtQuadrant.html>

or two in, depending on our pain threshold). We may even get to the stage where we consider a Big Rewrite.

Our advice is this. There are two practices that should be initiated from the beginning that will allow us to pay down technical debt later on: continuous integration and a small number of basic unit and user-journey tests. The moment a product or feature goes from being an experiment to validated, we need to start aggressively paying down technical debt—adding more user-journey tests, employing good architectural practices such as modularization, and making sure all new code written on the feature uses Test Driven Development (good engineers will already use TDD).

Having forced ourselves to do something that should be unnatural to engineers—hack out embarrassingly crappy code and get out of the building to get validation from early on—we must then pull the lever hard in the other direction, kill the momentum, and transition our focus from trying to build the right *thing*, to building it in the right *way*. Needless to say, this requires extreme discipline.

Choosing at what point in the lifecycle of our product or feature to pay down our technical debt is an art. If we find (as many do) that we’ve gone too far down the path of accumulating technical debt, we address alternatives to the Big Rewrite in (to come).

Engines Of Growth

In *The Lean Startup*, Eric Ries argued that there are three key strategies for growth—choose one.

- **Viral**—include any product that causes new customers to sign up as a necessary side-effect of existing customers’ normal usage: Facebook, MySpace, AIM/ICQ, Hotmail, Paypal. Key metrics are Acquisition and Referral, combined into the now-famous viral coefficient.
- **Pay**—we use a fraction of the lifetime value of each customer and flow that back into paid acquisition through SEM, banner ads, PR, affiliates, etc. The spread between your LTV and blended CPA determines either your profitability or your rate of growth, and a high valuation depends on balancing these two factors. Retention is the key goal in this model. Examples are Amazon and Netflix

- **Sticky** — something is causing their customers to become addicted to their product, and no matter how we acquire a new customer, we tend to keep them. The metric for sticky is “churn rate” — the fraction of customers in any period who fail to remain engaged with our product/service. This has led to exponential growth. For eBay, this is caused by the incredible network effects of their business.

However for enterprises there are further growth options to consider;

- **Expand** — built an adaptive initial business models that we could simply evolve and expand further by opening up new geographics and categories. Amazon has provided this excellently moving from selling books to an e-commerce stores offering new retail categories. The key with this growth strategy is the initial targeted market is large enough to support multiple phases of growth over time.
- **Platform** — once we have a successful core product, we transform it into a platform, around which an “ecosystem” of complementary products and services can be developed by both internal and external providers. Microsoft did this with Windows by creating MS Office, Money and other support packages, including those developed by external vendors — similarly Apple’s AppStore, Salesforce’s Force.com and Amazon’s marketplace.

Great products, tools and practices both internal and external, have always spread by word of mouth due to having a truly compelling value proposition and a brand that customers are proud to advocate. If our growth is a derivative of our customers, then it will happen without having to invest. If not, we will be limited by the effort required to manually service our customers.

Ultimately, our product is the key driver of growth. If we build a truly compelling solution that addresses a customer need, which they really love, they use it. More impressively, they become advocates and encourage others to use it — creating the best sales team we could wish to hope for to enable success.

Crossing the Horizon and Starting To Scale

In (to come), we highlighted that organizations must manage all three horizons concurrently. The ability to recongize, transition and convert initiatives through

these cycles holds the key for the future success, relevancy and longevity of the organization.

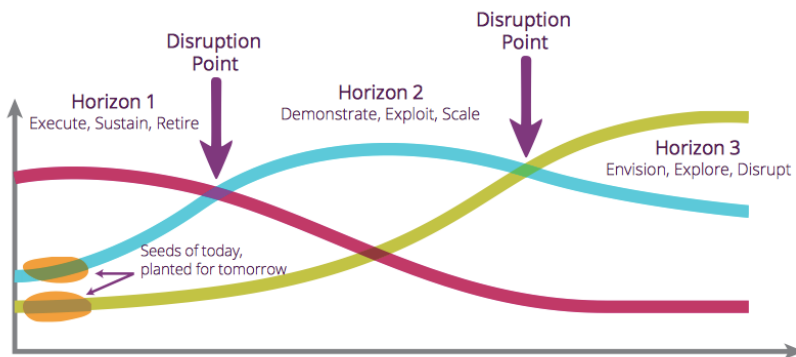


Figure 5-4. *The Three Innovation Horizons*

Counterintuitively, it is horizons two and three that need the most leadership support. Activities in horizon one tend to be fairly mature, well understood and focused on incremental improvement. Horizons two and three contain much more uncertainty and lower revenue, hence they can be easily crushed if not cared for. We must be aware of the pitfalls at each stage, including transitioning at the wrong time and selecting the wrong strategy for the next horizon. The key is to understand the opportunity and growth trajectory for the product.

When attempting to cross horizon three, leading indicators for customer satisfaction and continued engagement are important signals to monitor for future growth. Once we have found a customer, learnt how to address their needs and are confident of meeting their demand, we should seek to expand the customer market by geography, channel or offerings.

While exploring, we are testing the fit between product and market, typically through bespoke solutions for our initial customers. Exploiting is about finding an offering and business model that appeal to a broader customer set.

The five critical enablers of growth when moving from explore to exploit are;

- **Market** - it is imperative to select the right next market, ideally there are plenty of potential customers that will support our growth aspirations. We

must identify the elements that made us successful with our early adopter customers, then seek to find similar, larger groups to engage with. The insights we have learnt by working with our early adopters are key to informing this decision. Early adopters are also likely to spread their experiences with our product by word of mouth, eventually driving the product to *cross the chasm* through wider adoption.

- **Monetization model** - we must decide what is the best way to capture the value created by our offering as it essentially defines what will drive revenue for our business model. It is also difficult to change later.
- **Customer adoption** - how will we get customers on-board with the product? We must be careful not to make major concessions to any individual group across product and pricing to win over a large account. We must remain true to our product vision and manage the tension and demands of any single customers that could limit further growth.
- **Forget Big Bang launches** - support safety to continue to test and validate our product by containing the fire and working with smaller samples of customers. Build momentum by using alpha and beta products launches with targeted customer segments. As we gain more confidence, understanding and success, we boarden our customer base. Ideally we want customers coming to us with problems to solve and not resort to pushing new products on them.
- **Team engagement** - we must do all we can to keep the team together to protect culture, learning speed and acquired knowledge. We do not want to create an over-the-wall mentality between innovation and operation teams. Collaboration, directed towards organization learning and development, is key to making an innovation culture stick as we start to scale and onboard new team members.

Innovation over Management Accounting

Traditonal management accounting practices and operational measurement, discussed in (to come), are optimized to support horizon one and known entities. When considering horizon three and to a large extent horizon two, we need to focus on innovation *over* management accounting.

Pirate Metrics created by Dan McClure ¹⁰ are an elegant way to model any service-oriented business;

Table 5-2. *Pirate Metrics: AARRR!*

Name	Purpose	Example
Acquisition	Get people to find out about the product, either by referrals or paid for traffic	Word of mouth, Traffic, Mentions, Cost per Click, Cost of Acquisition
Activation	Get people engaged with the product and activated their account	Do <i>shoppers</i> sign up to start using the product? How many completed on-boarding process or used service once?
Retention	Understand if we have achieved product-market fit as people continually return to the service to use it	We retain a high percentage of customers once activated? How long since last visit, daily/month active users, churn rate?
Revenue	Generate cash from customer usage	Affiliate marketing, purchasing of services, customer life value, conversion rate, shopping cart size
Referral	Customer become brand advocates for our product, will it become viral?	Referring others customers to the product to start using it? What is our viral coefficient, viral cycle time

Other important measure to consider in relation to exiting explore are;

Table 5-3. *Explore to Exploit Growth Metrics*

Measure	Purpose	Exit signal
Customer Acquisition Cost	is the cost associated in convincing a customer to buy a product	Lower the better as we can get new customer cheaply
Viral coefficient	A quantitative measure of virality of the product	Higher the better as customers are engaging with the product
Customer Lifetime Value	The total value of a customer to the organization	Lower than the cost of customer acquisition
Monthly burn rate	The amount of money required to run the team, providing a runway for how long we can operate	We are generating enough revenue, along with funding to support the team existence

10. Pirate Metrics, <http://www.slideshare.net/dmc500hats/startup-metrics-for-pirates-long-version>

Innovation Takes Time - From Auction To Marketplace.

Amazon auctions (later known as zShops) was launched in March 1999, in response to the success of eBay. The site was promoted heavily from the home, category and individual product pages. Despite the promotion, one year after launch it had only achieved a 3.2% share of the online auction compared to 58% for eBay, and subsequently declined thereafter.

In November 2000, zShops was renamed to 'Amazon Marketplace' offering competitive prices on products available through third-party sellers along the standard product listings. The strategy, initially driven by the need to compete with eBay, was adjusted to align to Amazon's strategic focus on low pricing.

Extending the model further, Amazon introduced the sale of used products through the seller marketplace, providing another revenue stream without any impact to its supply chain. Advertising, packing and shipping are handled exclusively by sellers, with Amazon taking a cut of the transaction for providing the sales channel with minimal cost.

In 2012, Amazon's Marketplace service produced 12% of revenues ¹¹ with total unit sales increasing 32% from the previous year. ¹²

By creating an environment where we reconsider how we account, define and measure validated learning, we can start to measure and communicate if and when our initiatives are getting traction. By continually experimenting with our customers, to move our *One Metric That Matters* as cheaply and quickly as possible, we can limit our investment, reduce associated risks and maximize our learning. An evidence based approach to product development provides safety, context and cover to act for stakeholders and is a catalyst for change in the larger organization.

11. <http://vator.tv/news/2013-11-25-how-does-amazon-make-money>

12. <http://www.internetretailer.com/2013/01/29/amazon-sales-top-61-billion-2012>

Conclusion

In this chapter we have explained how to exit *explore* using the principles of innovation accounting combined with strategies for engines of growth to increase our customer based and start to *exploit* our validated solution.

We have identify the three key areas to consider during this stage;

- find a customers to act as a co-creator of value, using their feedback and needs to further experiment and refine our value proposition before aiming for a wider market
- continue to focus on learning over revenue by taking a narrow customer focus, doing things that do not scale to validate each assumptions we have about our solution. We do not need to build requirements, we need to answer questions about the desired functionality of our product. We do this by staying close to the customer and not automating our touch-points away simply to scale our solution
- focus on improving customer engagement over quick financial gain — with more satisfied customers will come revenue. As we improve our understanding of our customer and product opportunity, we will have the best insight to base our decision on what monetization model will insure the ongoing success for the product

Our next goal is to focus on growing to a point of break-even for the operation of the team, and the revenue from our customers. In (to come) we will share how to start to build the product right, now that we have confidence that we are building the right product.

Questions for readers:

- What would be on a scorecard of customer and business metrics that matter to you - decide on a One Metric That Matter depending on the stage your product is it? Formulate experiments that focus on improving it.
- Create an adoption curve in your organization and identify and map the key stakeholder and their influence for each stage of the curve. Formulate strategies and specific points when and how to engage them.

- Think how you can regularly showcase your solution and scorecard to obtain alignment, buy-in and funding from key stakeholders and supporting departments to not get crushed.
- Use your Story Map to create a runway of experiments to continue to test and validate your business model hypothesis with customers.
- Create an Impact and Effort matrix ¹³ to organize your experiments and optimize for cycle time of testing with your identified market.

13. Impact & Effort Matrix, <http://www.gamestorming.com/games-for-decision-making/impact-effort-matrix-2/>